Introduction to Probability and Statistics

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Final Assignment

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Tutorial-01

Q1:
A sample of 26 offshore oil workers took part in a simulated escape exercise, resulting in the accompanying data on time (sec) to complete the escape:

325, 325, 334, 339, 356, 356, 359, 359, 363, 364, 364, 366, 369, 370, 373, 373, 374, 375, 389, 392, 393, 394, 397, 402, 403, 424.

(a) Construct a stem and leaf display of the data. How does it suggest that the sample mean and median will compare?

(b) Calculate the values of the sample mean and median.

(c) By how much could the largest time, currently 424, be increased without affecting the value of the sample median? By how much could this value be decreased without affecting the value of the sample median?

(d) What are the values of Trimmed mean, Trimean and Interquartile mean when the observations are re-expressed in minutes?

($\alpha = 10\%$)

Q1) Ans

a) The stem and leaf plot of the data:

Stem	Leaf
32	5, 5
33	4, 9
34	6, 6, 9, 9
36	3, 4, 4, 6, 9
37	0, 3, 3, 4, 5
38	9
39	2, 3, 4, 7
40	2, 3
42	4

b

Here, $n = 26$

$\therefore$ Sample mean, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

$= \frac{1}{26}(325 + 325 + 334 + \cdots + 402 + 403 + 424)$

$= \frac{9638}{26}$

$= 370.692$

$Me = \frac{1}{2}\left[x_{n/2} + x_{(\frac{n}{2}+1)}\right]$

$= \frac{1}{2}\left[x_{26/2} + x_{(26/2+1)}\right]$

$= \frac{1}{2}\left[x_{13} + x_{14}\right] = \frac{1}{2}(369 + 370) = 369.5$

c

The last value of the given data set, currently 424, can be increased any number of time upto infinity. It doesn't affect the value of the sample median.

Here, the median value depends on x_{13} and x_{14}. Those are 369 and 370 respectively. So, we can decrease the value 424 upto 370. If the value 424 changes to less than 370, it will affect the value of the median..

d

10% trimmed mean, $\bar{x}_{tr(10)}$

$= (339 + 356 + 356 + 359 + \ldots\ldots + 393 + 394 + 397)/20$

$= \frac{7425}{20} = 371.25 = 6.1875$ mins

Trimean

$Q_i = \frac{1}{2}\left[x_{\frac{in}{4}} + x_{\left(\frac{in}{4}+1\right)}\right]$

$Q_1 = \frac{1}{2}[x_6 + x_7]$

$= \frac{1}{2}(356 + 359) = 357.5$

$Q_2 = \frac{1}{2}(x_{13} + x_{14}) = 369.5$

$$Q_3 = \frac{1}{2}(x_{19} + x_{20})$$
$$= \frac{1}{2}(389 + 392)$$
$$= 390.5$$

$$\therefore TM = \frac{Q_1 + 2Q_2 + Q_3}{4} = \frac{357.5 + (2 \times 369.5) + 390.5}{4}$$
$$= 371.75 = 6.195 \text{ mins.}$$

Interquartile Mean

$$\bar{x}_{IQM} = \frac{2}{n} \sum_{i=\frac{n}{4}+1}^{3n/4} x_i$$

$$= \frac{2}{26}(359 + 359 + \cdots + 373 + 389)$$
$$= \frac{1}{13} \times 4799$$
$$= 369.076$$
$$= 6.1513 \text{ mins.}$$

Q2

A certain type of refrigerator comes in three sizes, each of which is available with or without an ice maker. The accompanying table gives the relevant probabilities for a randomly selected purchaser:

		Size		
		17.5ft³	21 ft³	24.5 ft³
Icemaker	Yes	0.12	0.17	0.24
	NO	0.18	0.13	0.16

(i) What is the probability that the purchaser selected a 21-ft^3 refrigerator? A refrigerator with an icemaker?

(ii) Given that the purchaser select a model with an icemaker, what is the probability of selecting a 21 ft^3? How does this compare to the unconditional probability of selecting a 21-ft^3 model?

(iii) Given that the purchaser selected a model with atleast 20-ft^3 of space, what is the probability that the selected refrigerator did not have an ice-maker?

(i)

Suppose
A = Selected a 21 ft^3 refrigerator
B = Selected a refrigerator with an icemaker

$\therefore P(A) = 0.17 + 0.13 = 0.30$
$P(B) = 0.12 + 0.17 + 0.24 = 0.53$

(ii)

Given that the purchaser select a model with an icemaker, the probability that it was 21-ft^3

$$= \frac{0.17}{0.12 + 0.17 + 0.24} = 0.321$$

On the other hand, the unconditional probability of selecting a 21 ft³ model = 0.3 which is less than the conditional probablety.

(iii)

A = Selected a model atleast 20-ft³
B = Selected a refrigerator didnot have an ice maker.

∴ the probablity = $\frac{P(A \cap B)}{P(A)}$

$= \frac{0.13 + 0.16}{0.17 + 0.13 + 0.24 + 0.16}$

$= 0.4143$

Q3

An insurance Company offers its policy holders a number of different premium payment options. For a randomly selected policy holder, Let X = the number of months between successive payments. The cdf of X is as follows:

$$F_X(x) = \begin{cases} 0 & x < 1 \\ 0.30 & 1 \le x < 3 \\ 0.40 & 3 \le x < 4 \\ 0.45 & 4 \le x < 6 \\ 0.60 & 6 \le x < 12 \\ 1 & 12 \le x \end{cases}$$

(i) What is the pmf of X?
(ii) Using just the cdf, compute $P[3 \leq x \leq 10]$ and $P[4 \leq x]$

(i)

X	$F_X(x)$	Pmf, $P(x)$
$x < 1$	0	0
$1 \leq x < 3$	0.30	0.30
$3 \leq x < 4$	0.40	0.10
$4 \leq x < 6$	0.45	0.05
$6 \leq x < 12$	0.60	0.15
$12 \leq x$	1	0.4

We use, $P(x_i) = F(x_i) - F(x_{i-1})$

(ii)

$P[3 \leq X \leq 10] = F(10) - F(2)$
$= 0.60 - 0.30 = 0.30$

$P[4 \leq x] = 1 - F(3) = 1 - 0.40 = 0.60$

Q1

Define Scatter Diagram, Rank correlation, Correlation and Regression analysis. State the characteristics of Karl Pearson's Correlation co-efficient. Comment on the following : (i) $r = 0.95$ (ii) $r = 0$ (iii) $r = 0.90$ (iv) $r = 0.62$

⇒ Scatter Diagram: A scatter Diagram also known as scatter plot is a tool for analyzing relationships between two variables for determining how closely the two variables are related.

Rank correlation: A rank correlation co-efficient measures the degree of similarity between two rankings.

Correlation and regression analysis:

A correlation analysis provides information on the strength and direction of the linear relationship between two variables, while a linear regression analysis estimates parameters in a linear equation that can be used to predict values of one variable based on the other.

Correlation Co-efficient:

The process of measuring the linear relationship of two variables is known as correlation analysis. The amount of linear relationship is known as correlation coefficient. It is denoted by r and the value of r lies between -1 to 1.

Three types of correlation coefficient:

1) Positive correlation: If X and Y have a strong positive linear correlation, r is close to +1

2) Negative correlation: r is close to -1. It indicates a relationship between X and Y such that as X increases, Y decreases.

3) Spurious correlation: r is close to 0. A value near zero means there is a random, non-linear relationship between X and Y.

(i) $r = 0.95$

Here, $0.80 \leq |r| \leq 1.00$ indicates a very strong correlation.

(ii) $r = 0$
there is no correlation.

(iii) $r = -0.90$
$|r| = 0.90$ lies between 0.80 - 1.00 that indicates a very strong correlation.

(iv) $r = 0.67$; $|r|$ lies between 0.60 - 0.79 that indicates strong correlation.

Q5.

A bank operate both drive up facility and a walk-up window. On a randomly selected day X = the proportion of time that the drive up facility is in use (at least one customer is being served or waiting to be served). and Y = the proportion of time that the walk up window is in use. Then the set of possible values for (X, Y) is the rectangle $D = \{(x,y) : 0 \le x \le 1, 0 \le y \le 1\}$. Suppose the joint density function of (X, Y) is given by $f_{XY}(x,y) = \frac{6}{5}(x+y^2)$; $0 \le x \le 1, 0 \le y \le 1$

(i) Verify that $f_{XY}(x,y)$ is a legitimate pdf.

(ii) What is the probablity that neither facility is busy more than one-quarter of the time?

(iii) Find the marginal pdf of X and marginal pdf of Y.

(i)

Given that,

$f_{xy}(x,y) = \frac{6}{5}(x+y^2)$; $0 \le x \le 1$, $0 \le y \le 1$

$$\int_0^1 \int_0^1 \frac{6}{5}(x+y^2)\,dx\,dy$$

$$= \frac{6}{5}\int_0^1 \left[\frac{x^2}{2} + xy^2\right]_0^1 dy$$

$$= \frac{6}{5}\int_0^1 \left[\frac{1}{2} + y^2\right] dy$$

$$= \frac{6}{5}\left[\frac{y}{2} + \frac{y^3}{3}\right]_0^1$$

$$= \frac{6}{5}\left[\frac{1}{2} + \frac{1}{3}\right]$$

$$= \frac{6}{5} \times \frac{5}{6} = 1$$

$\therefore f_{xy}(x,y)$ is a legitimate pdf. (verified)

(ii)

$$P\left(x \le \frac{1}{4}, y \le \frac{1}{4}\right) = \int_0^{1/4}\int_0^{1/4} f_{xy}(x,y)\,dx\,dy$$

$$= \int_0^{1/4}\int_0^{1/4} \frac{6}{5}(x+y^2)\,dx\,dy$$

$$= \frac{6}{5}\int_0^{1/4}\left[\frac{x^2}{2} + xy^2\right]_0^{1/4} dy$$

$$= \frac{6}{5}\int_0^{1/4}\left[\frac{1}{32} + \frac{y^2}{4}\right] dy$$

$$= \frac{6}{5}\left[\frac{y}{32} + \frac{y^3}{12}\right]_0^{1/4}$$

$$= \frac{6}{5}\left[\frac{1}{32\times 4} + \frac{1}{12\times 4^3}\right) = \frac{6}{5} \times \left(\frac{1}{128} + \frac{1}{768}\right)$$

$$= \frac{7}{640} \text{ (Ans)}$$

(iii)

Marginal pdf of X,

$$f_X(x) = g(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$

$$= \int_0^1 \frac{6}{5}(x+y^2)\,dy$$

$$= \frac{6}{5}\left[xy + \frac{y^3}{3}\right]_0^1$$

$$= \frac{6}{5}\left(x + \frac{1}{3}\right)$$

$$= \frac{6x}{5} + \frac{2}{5}, \quad 0 < x \leq 1$$

Marginal pdf of Y,

$$f_Y(y) = h(y) = \int_0^1 \frac{6}{5}(x+y^2)\,dx$$

$$= \frac{6}{5}\left[\frac{x^2}{2} + xy^2\right]_0^1$$

$$= \frac{6}{5}\left(\frac{1}{2} + y^2\right)$$

$$= \frac{3}{5} + \frac{6}{5}y^2, \quad 0 \leq y \leq 1$$

Tutorial - 02

Q1:
Define Gamma distribution and Weibull distribution. Write down some application of Gamma distribution.

⇒ Gama function,

$$\Gamma(n) = \int_0^\infty x^{n-1} e^{-x} dx \quad \text{for } n>0$$

Gama ~~probability densi~~ distribution is a family of right-skewed, continuous probability distribution

$$f(x) = \frac{\lambda}{\Gamma\kappa} (\lambda x)^{\kappa-1} e^{-\lambda x}, \quad x>0$$
$$= 0 \quad ; \text{otherwise}$$

Here, κ = shape parameter, λ = scale parameter.

Weibull Distribution is a generalized gamma distribution with both shape parameters equal to K.

Pdf is,

$$f(x) = \frac{\beta}{\delta}\left(\frac{x-\gamma}{\delta}\right)^{\beta-1} \exp\left[-\left(\frac{x-\gamma}{\delta}\right)^{\beta}\right], \quad x \geq \gamma$$
$$= 0, \quad \text{otherwise.}$$

Applications of Gamma Distribution:

i) Modeling size of insurance claims, rainfall
ii) Errors in multilevel Poisson regression model
iii) Wireless communication
iv) Genomics.

Q2

What do you mean by Binomial variate? Suppose X follows Binomial distribution with parameter n and p then show that,

(i) $E\left[\frac{X}{n} - p\right]^2 = \frac{pq}{n}$

(ii) $cov\left[\frac{X}{n}, \frac{n-X}{n}\right] = \frac{-pq}{n}$

⇒ In probability theory and statistics, the binomial distribution with parameters n and p is the discrete probability distribut of the number of successes in a sequence of n independent experiments, each asking a yes-no question, and each with its own Boolean-valued outcome: success or failure,

(i) $E\left[\frac{X}{n}-p\right]^2$

$= E\left[\frac{X^2}{n^2} - 2\cdot\frac{X}{n}\cdot p + p^2\right]$

$= \frac{1}{n^2}E[X^2] - 2\cdot\frac{p}{n}E[X] + E[p^2]$

$= \frac{1}{n^2}E[X(X-1)+X] - \frac{2p}{n}\cdot np + p^2$

$= \frac{1}{n^2}[n(n-1)p^2 + np] - 2p^2 + p^2$

$= \frac{n^2p^2 - np^2 + np}{n^2} - p^2$

$= p^2 - \frac{p^2}{n} + \frac{p}{n} - p^2$

$= \frac{p}{n}(1-p)$

$= \frac{pq}{n}$

[Showed]

(ii) $cov[X,Y] = E[XY] - E[X]\cdot E[Y]$

$\therefore cov\left[\frac{X}{n}, \frac{n-X}{n}\right]$

$= E\left[\frac{X}{n}\cdot\frac{n-X}{n}\right] - E\left[\frac{X}{n}\right]\cdot E\left[\frac{n-X}{n}\right]$

$= \frac{1}{n^2}E[nX - X^2] - \frac{1}{n}E[X]\cdot\frac{1}{n}E[n-X]$

$= \frac{1}{n^2}\left[nE[X] - E[X(X-1)+X]\right] - \frac{1}{n^2}\cdot np(n-np)$

$= \frac{1}{n^2}\left[n\cdot np - n(n-1)p^2 - np\right] - \frac{n^2p - n^2p^2}{n^2}$

$= \frac{n^2p - n^2p^2 + np^2 - np}{n^2} - \frac{n^2p - n^2p^2}{n^2}$

$$= p - p^2 + \frac{p^2}{n} - \frac{p}{n} - p + p^2$$

$$= -\frac{p}{n}(1-p)$$

$$= \frac{-pq}{n}$$ [showed]

Q3:

A particular telephone number is used to receive both voice calls and fax messages. Suppose that 30% of the incoming calls involve fax messages and consider a sample of 40 incoming calls. What is the probablity that -

(a) At most six of the calls involve a fax messages?

(b) Exactly six of the calls involve a fax messages?

(c) At least six of the calls involve a fax messages?

(d) More than six of the calls involve a fax messages?

$$(a)\ P(X \le 6) = \sum_{x=0}^{6} f(x)$$

$$= \sum_{x=0}^{6} \binom{n}{x} (0.3)^{x} (0.7)^{n-x}$$

$$= \sum_{x=0}^{6} \binom{40}{x} (0.3)^{x} (0.7)^{40-x}$$

$$= 0.0237$$

$n = 40$

$P(fx) = 30\% = 0.30$

$P(\bar{x}) = (1-0.3) = 0.7$

06

(b) $P(X=6) = \binom{25}{6}\binom{40}{6}(0.3)^6(0.7)^{40-6}$

$= 0.015514$

(c) $P(X \geq 6)$

$= 1 - P(X \leq 5)$

$= 1 - \sum_{x=0}^{5} \binom{40}{x}(0.3)^x(0.7)^{40-x}$

$= 1 - 0.008618$

$= 0.991382$

(d) $P(X > 6)$

$= 1 - P(X \leq 6)$

$= 1 - 0.02307$

$= 0.9769$

Q4

Consider writing onto a computer disk and then sending it through a certifier that counts the number of missing pulses. Suppose this number X has a Poisson distribution with parameter $\lambda = 0.2$.

(a) What is the probablity that a disk has exactly one missing pulse?

(b) What is the probablity that a disk has ~~exactly~~ at least two missing pulse?

(c) If two disks are independently selected what is the probablity that neither contains a missing pulse?

⇒(a) According to the given question, a computer disk and then sending it through a certifier that counts the number of missing pulses. Suppose this number X has a poisson distribution with parameter $\lambda = 0.2$

Therefore X follows Poison distribution with the probablity mass function,

$$f(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \qquad x = 0, 1, 2, 3, \cdots \infty$$

Therefore the probablity that a disk has

exactly one missing pulse is determined as:

$$P(x=1) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$= \frac{e^{-0.2}(0.2)^{1}}{1!}$$

$$= \cancel{0.0735}\ 0.1637$$

(b) The probablity that a disk has at least two missing pulse:

$$P(x \geq 2) = 1 - [P(x=0) + P(x=1)]$$

$$= 1 - \left[\frac{e^{-0.2}\cdot(0.2)^{0}}{0!} + \frac{e^{-0.2}\times(0.2)^{1}}{1!}\right]$$

$$= 1 - [0.8187 + 0.1637]$$

$$= 0.0176$$

(c) Probablity that first disk has no missing pulse,

$$P(x=0) = \frac{e^{-0.2}\times(0.2)^{0}}{0!} = 0.8187$$

Probabity that the second disk also has no missing pulse,

$$P(x=0) = \frac{e^{-0.2}\times(0.2)^{0}}{0!} = 0.8187$$

Therefore, if the two disks are independently selected, the probablity that neither contains a missing pulse is $0.8187 \times 0.8187 = 0.67026$

(Ans.)

Q5
Find Mean, Variance and moment generating function of Exponential distribution.

⇒ Exponential distribution function,

$$f(x,\lambda) = \lambda e^{-\lambda x} \qquad x>0$$

$$\text{Mean, } E(x) = \int_0^\infty x f(x)\,dx$$

$$= \int_0^\infty x \cdot \lambda e^{-\lambda x}\,dx$$

$$= \lambda \int_0^\infty x e^{-\lambda x}\,dx$$

$$= \lambda \left\{ \left[x \cdot \frac{e^{-\lambda x}}{-\lambda} \right]_0^\infty - \int_0^\infty 1 \cdot \frac{e^{-\lambda x}}{-\lambda}\,dx \right\}$$

$$= \lambda \{0+0\} + \frac{1}{\lambda} \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^\infty$$

$$= -\frac{d\lambda}{\lambda^2} \left[e^{-\lambda x} \right]_0^\infty$$

$$= -\frac{1}{\lambda^\infty} \left[0 - 1 \right]$$

$$= \frac{1}{\lambda}$$

$$V(x) = E(x^2) - [E(x)]^2$$

$$= \int_0^\infty x^2 \cdot \lambda e^{-\lambda x}\, dx - \left(\frac{1}{\lambda}\right)^2$$

$$= \lambda \left[\left(\frac{x^2 e^{-\lambda x}}{-\lambda} \right)_0^\infty - \int_0^\infty 2x \cdot \frac{e^{-\lambda x}}{-\lambda}\, dx \right] - \frac{1}{\lambda^2}$$

$$= 0 - \lambda \cdot 2 \cdot \frac{1}{-\lambda} \int_0^\infty x e^{-\lambda x}\, dx - \frac{1}{\lambda^2}$$

$$= 2 \cdot \frac{1}{\lambda^2} - \frac{1}{\lambda^2}$$

$$= \frac{1}{\lambda^2}$$

$$M_x(t) = E(e^{tx})$$

$$= \int_0^\infty e^{tx} \cdot \lambda e^{-\lambda x}\, dx$$

$$= \lambda \int_0^\infty e^{-x(\lambda - t)}\, dx$$

$$= \lambda \left[\frac{e^{-x(\lambda - t)}}{-(\lambda - t)} \right]_0^\infty$$

$$= \frac{\lambda}{\lambda - t} \cdot 1$$

$$= \frac{1}{\frac{\lambda - t}{\lambda}}$$

$$= \frac{1}{1 - \frac{t}{\lambda}} = \left(1 - \frac{t}{\lambda}\right)^{-1}$$

(Ans.)

Assignment
Exercise-01

20.02.2021

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Variables	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
Weight (kg)	36	37	40	41	41	44	44	45	47

x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{16}	x_{17}	x_{18}
48	48	50	51	51	51	51	51	52

x_{19}	x_{20}	x_{21}	x_{22}	x_{23}	x_{24}	x_{25}	x_{26}	x_{27}
55	55	55	55	57.5	59	60	60	64

x_{28}	x_{29}	x_{30}	x_{31}	x_{32}	x_{33}	x_{34}	x_{35}	x_{36}
65	67	68	70	75	75	78	78	87

Arithmetic Mean, $AM = \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

$$= \frac{x_1 + x_2 + x_3 + \cdots\cdots + x_{36}}{36}$$

$$= \frac{36 + 37 + 40 + \cdots\cdots + 87}{36}$$

$$= \frac{2011.5}{36}$$

$$= 55.875 \quad \text{(Ans.)}$$

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Geometric Mean, $GM = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}}$

$= \sqrt[n]{x_1 \times x_2 \times \cdots \times x_{36}}$

$= (x_1 \times x_2 \times x_3 \times \cdots \times x_{35} \times x_{36})^{\frac{1}{36}}$

$= (36 \times 37 \times 40 \times \cdots \times 78 \times 87)^{\frac{1}{36}}$

$= \text{Antilog} \frac{1}{n} \sum_{i=1}^{n} \log_{10}(x_i)$

$= \text{Antilog} \frac{1}{36} [\log_{10}(36) + \log_{10}(37) + \cdots + \log_{10}(87)]$

$= \text{Antilog} \frac{1}{36} (62.52453354)$

$= 54.5497$ (Ans.)

Median, $Me = \frac{1}{2} \left[x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}\right]$ [As n is even]

$= \frac{1}{2} \left[x_{\left(\frac{36}{2}\right)} + x_{\left(\frac{36}{2}+1\right)}\right]$

$= \frac{1}{2} [x_{18} + x_{19}]$

$= \frac{1}{2} [52 + 55]$

$= 53.5$ (Ans.)

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Mode:

Mode is the value of the variable which occurs most frequently in the series of the observation of the variable.

Here 51 occurs most frequently (5 times).

So, mode is 51.

(Ans:)

Trimmed Mean: (Considering 10% trimmed mean)

Trimmed Mean Percent $= \frac{10}{100} = 0.1$

Sample size, $n = 36$.

Trimmed count, $g =$ ~~Floor~~ (Trimmed Mean Percent X Sample size)

$= $ ~~Floor~~ (0.1×36)

$= 3.6 \simeq 4$

So, we remove ~~first~~ 4 numbers from both end and beginning.

∴ First 4 numbers 36, 37 and, 40, 41 and last 4 numbers 75, 78, 78, 87 are removed.

∴ Trimmed Mean, $\bar{x}_{tr(10)} = \frac{1}{n-8} \sum_{i=5}^{32} x_i$

$= \frac{1}{36-8}(41+41+44+\ldots\ldots+70+75+75)$

$= \frac{1}{28} \times 1539.5$

$= 54.982$ (Ans.)

Winsorized Mean: (considering 10%):

10% of 36 $= \frac{36 \times 10}{100} = 3.6 \cong 4$

$$\therefore WM = \frac{x_5+x_5+x_5+x_5+x_5+x_6+\ldots\ldots+x_{30}+x_{31}+x_{32}+x_{32}+x_{32}+x_{32}+x_{32}}{n}$$

$= \frac{2003.5}{36}$

$= 55.653$ (Ans.)

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Interquartile mean:

$$\bar{x}_{IQM} = \frac{2}{n} \sum_{i=\left(\frac{n}{4}+1\right)}^{\frac{3n}{4}} x_i$$

$$= \frac{2}{36} \sum_{i=\left(\frac{36}{4}+1\right)}^{\frac{3\times 36}{4}} x_i$$

$$= \frac{1}{18} \sum_{i=10}^{27} x_i$$

$$= \frac{1}{18}\left(x_{10} + x_{11} + \cdots + x_{26} + x_{27}\right)$$

$$= \frac{1}{18}\left(48 + 48 + \cdots + 60 + 64\right)$$

$$= \frac{973.5}{18}$$

$$= 54.083$$

(Ans.)

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Quartile:

$$Q_i = \frac{1}{2}\left[x_{\left(\frac{in}{4}\right)} + x_{\left(\frac{in}{4}+1\right)}\right] \quad [\because n \text{ is even}]$$

$$\therefore Q_1 = \frac{1}{2}\left[x_{\left(\frac{1\times 36}{4}\right)} + x_{\left(\frac{1\times 36}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_9 + x_{10}\right]$$

$$= \frac{1}{2}\left[47 + 48\right]$$

$$= 47.5 \text{ (Ans.)}$$

$$Q_2 = \frac{1}{2}\left[x_{\left(\frac{2\times 36}{4}\right)} + x_{\left(\frac{2\times 36}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{18} + x_{19}\right]$$

$$= \frac{1}{2}\left[52 + 55\right]$$

$$= 53.5 \text{ (Ans.)}$$

$$Q_3 = \frac{1}{2}\left[x_{\left(\frac{3\times 36}{4}\right)} + x_{\left(\frac{3\times 36}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{27} + x_{28}\right]$$

$$= \frac{1}{2}\left[64 + 65\right]$$

$$= 64.5 \text{ (Ans.)}$$

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Inter Quartile Range, $IQR = Q_3 - Q_1$

$= 64.5 - 47.5$

$= 17$ (Ans.)

Trimean, $TM = \frac{Q_1 + 2Q_2 + Q_3}{4}$

$= \frac{47.5 + 2 \times 53.5 + 64.5}{4}$

$= 54.75$

(Ans.)

Percentile; $P_i = \frac{1}{2}\left[x_{\left(\frac{i \times n}{100}\right)} + x_{\left(\frac{i \times n}{100}+1\right)}\right]$ [n is even]

$\therefore P_{60} = \frac{1}{2}\left[x_{\left(\frac{60 \times 36}{100}\right)} + x_{\left(\frac{60 \times 36}{100}+1\right)}\right]$

$= \frac{1}{2}\left[x_{(21.6)} + x_{(22.16)}\right]$

$= \frac{1}{2}\left[x_{(21)} + x_{(22)}\right]$

$= \frac{1}{2}[55 + 55]$

$= 55$ (Ans.)

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Assignment 02

04.03.21

Iffat Ara Sanzida
Roll: 344

Given Data Set including my weight (51kg) is:
36, 37, 40, 41, 41, 44, 44, 45, 47, 48, 48, 50, 51, 51, 51, 51, 51, 52, 55, 55, 55, 55, 58, 59, 60, 60, 64, 65, 67, 68, 70, 75, 75, 78, 78, 87.

1. Stem and Leaf Plot:

Stem	Leaf
3	6, 7
4	0, 1, 1, 4, 4, 5, 7, 8, 8
5	0, 1, 1, 1, 1, 1, 2, 5, 5, 5, 5, 8, 9
6	0, 0, 4, 5, 7, 8,
7	0, 5, 5, 8, 8
8	7

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Frequency Histogram & Relative frequency Histogram from the frequency table mentioned below:

Minimum = 36, Maximum = 87

Range, R = Maximum - Minimum

$= 87 - 36$

$= 51$

Sturge's formula for number of class,

$K = 1 + 3.332 \log_{10}(n)$

$= 1 + 3.332 \log_{10}(36)$

$= 6.185$

≈ 7

Class width, $h = \frac{R}{K}$

$= \frac{51}{7}$

$= 7.28$

≈ 8

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Class Interval	Frequency	Cumulative Frequency	Relative Frequency	Cumulative Relative Frequency
36-43	5	5	0.139	0.139
44-51	12	17	0.333	0.472
52-59	7	24	0.194	0.667
60-67	5	29	0.139	0.806
68-75	4	33	0.111	0.917
76-83	2	35	0.056	0.972
84-91	1	36	0.028	1

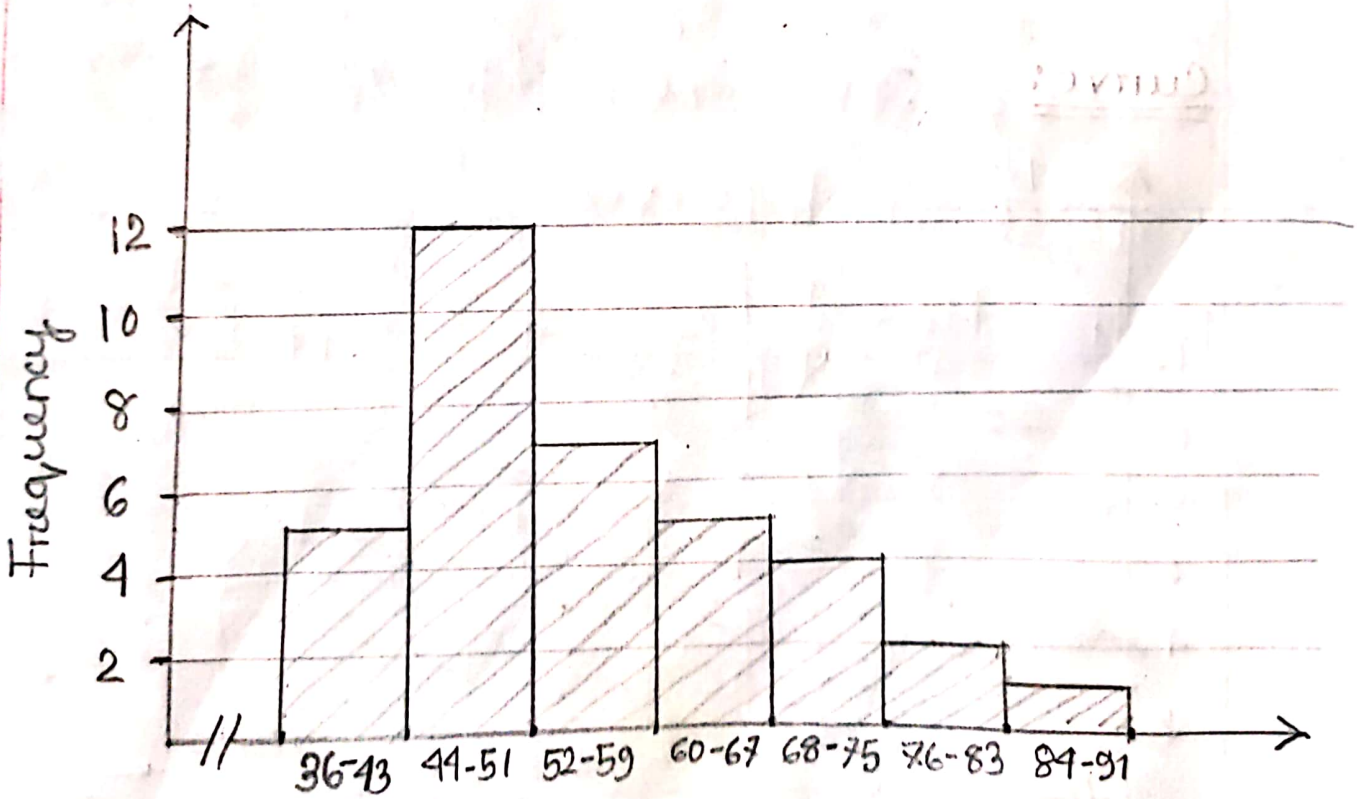

Fig: Frequency Histogram

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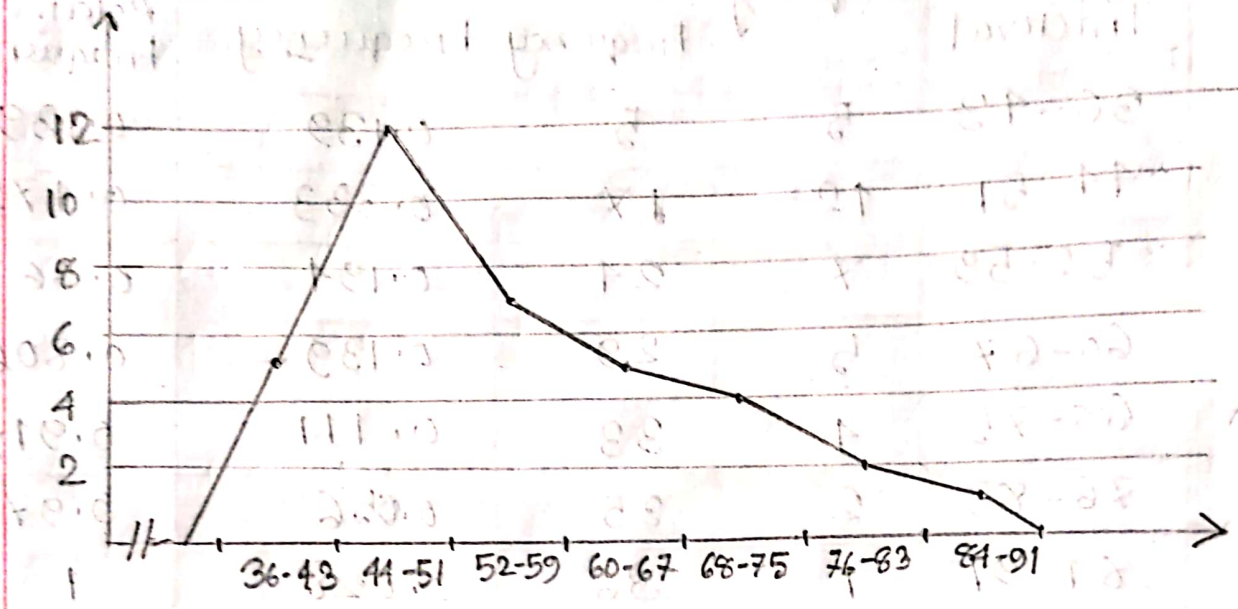
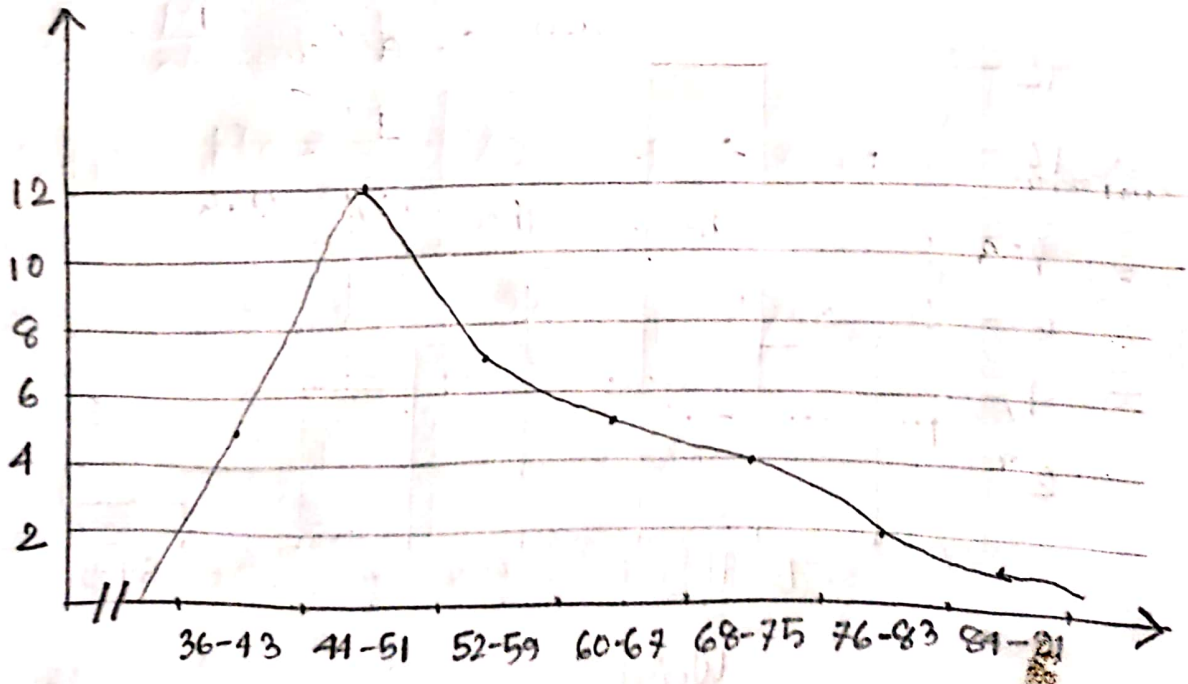
Polygon:



Curve:



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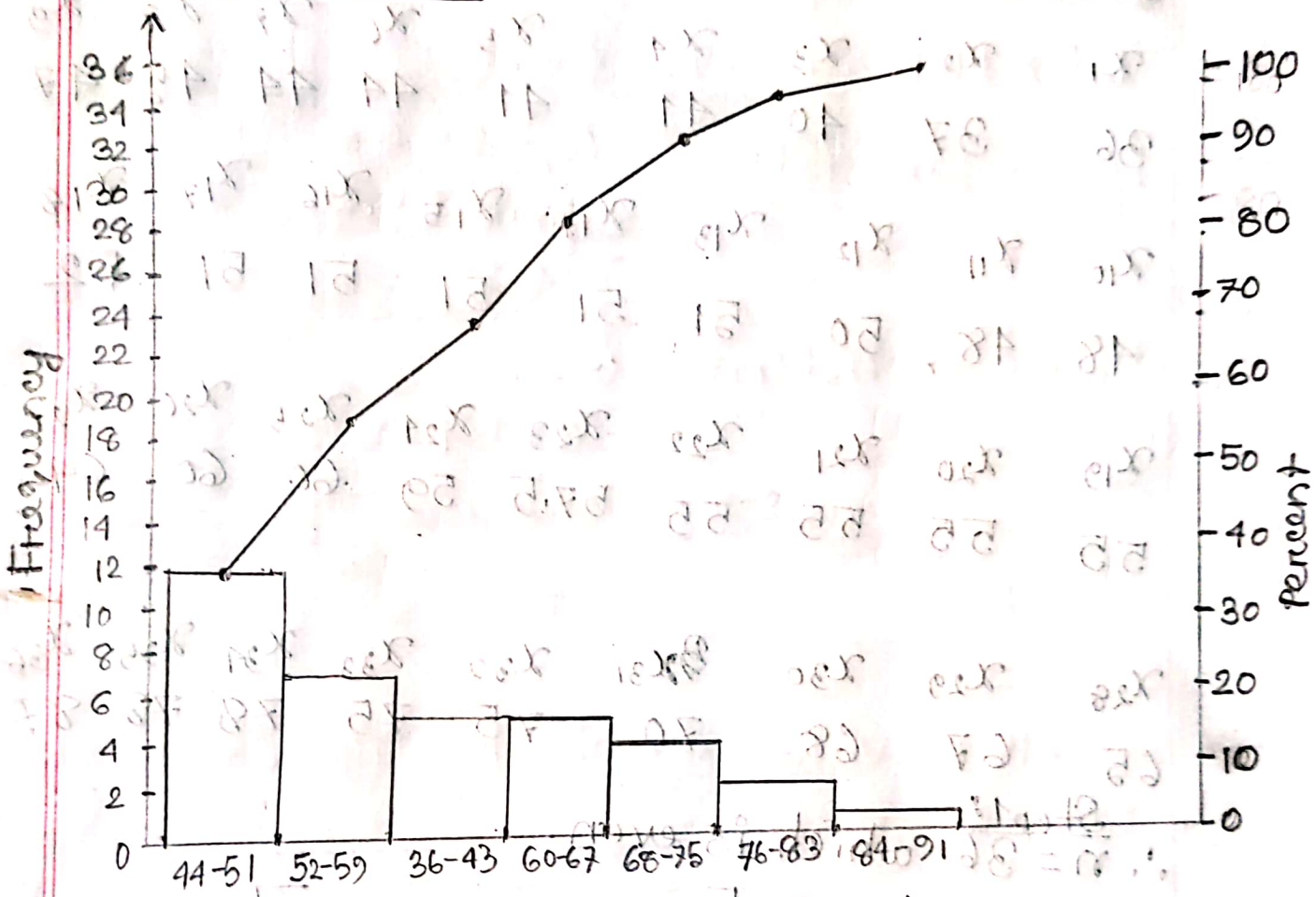
Pareto Chart:



Frequency:	12	7	5	5	4	2	1
Percent:	33.3	19.4	13.9	13.9	11.1	5.6	2.8
Cum%:	33.3	52.7	66.6	80.5	91.6	97.2	100.0
Cumulative Frequency:	12	19	24	29	33	35	36

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Box and Whishker Plot:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
36	37	40	41	41	44	44	45	47

x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{16}	x_{17}	x_{18}
48	48	50	51	51	51	51	51	52

x_{19}	x_{20}	x_{21}	x_{22}	x_{23}	x_{24}	x_{25}	x_{26}	x_{27}
55	55	55	55	57.5	59	60	60	64

x_{28}	x_{29}	x_{30}	x_{31}	x_{32}	x_{33}	x_{34}	x_{35}	x_{36}
65	67	68	70	75	75	78	78	87

Step 1:

$\therefore n = 36$ which is even.

$$\text{Median, } Me = \frac{1}{2}\left[x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{\left(\frac{36}{2}\right)} + x_{\left(\frac{36}{2}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{18} + x_{19}\right]$$

$$= \frac{1}{2}\left[52 + 55\right]$$

$$= 53.5$$

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$$Q_1 = \frac{1}{2}\left[x_{\left(\frac{in}{4}\right)} + x_{\left(\frac{in}{4}+1\right)}\right]$$

$$\therefore Q_1 = \frac{1}{2}\left[x_{\left(\frac{1\times 36}{4}\right)} + x_{\left(\frac{1\times 36}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_9 + x_{10}\right]$$

$$= \frac{1}{2}\left[47 + 48\right]$$

$$= 47.5$$

$$Q_3 = \frac{1}{2}\left[x_{\left(\frac{3\times 36}{4}\right)} + x_{\left(\frac{3\times 36}{4}+1\right)}\right]$$

$$= \frac{1}{2}\left[x_{27} + x_{28}\right]$$

$$= \frac{1}{2}\left[64 + 65\right]$$

$$= 64.5$$

$$IQR = Q_3 - Q_1 = 64.5 - 47.5$$

$$= 17$$

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Step 2:

$1.5 \times IQR = 1.5 \times 17 = 25.5$

Lower Innerfence $= Q_1 - 25.5$

$= 47.5 - 25.5$

$= 22$

Upper Innerfence $= Q_3 + 25.5$

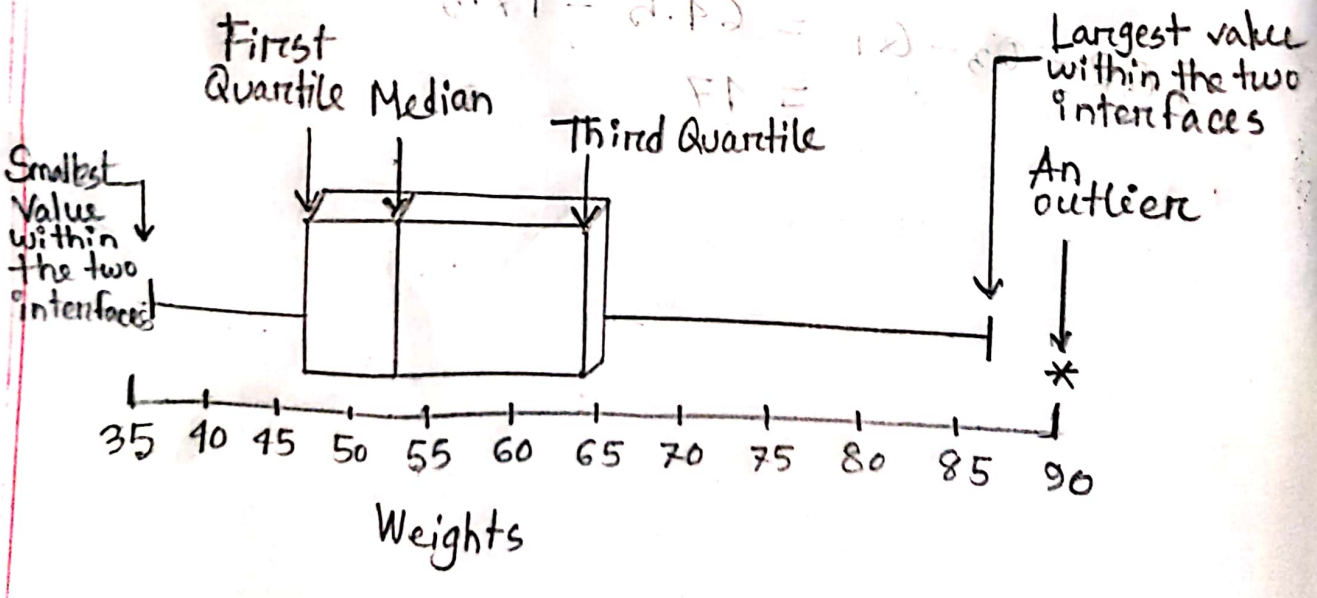
$= 64.5 + 25.5$

$= 90$

Step3:

Smallest value within two inner fences = 36

Largest " " " " = 87



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Sample variance, $S^2 = \frac{\Sigma(x_i - \bar{x})^2}{n-1}$

Here, Mean, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

$$= \frac{x_1 + x_2 + x_3 + \cdots + x_{36}}{36}$$

$$= \frac{36 + 37 + 40 + \cdots + 87}{36}$$

$$= 55.875$$

$$\therefore S^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_{36} - \bar{x})^2}{36-1}$$

$$= \frac{(36-55.875)^2 + (37-55.875)^2 + \cdots + (87-55.875)^2}{35}$$

$$= \frac{5565.6875}{35}$$

$$= 159.0196429$$

$\therefore$ Standard deviation, $\sigma = \sqrt{S^2}$

$$= \sqrt{159.0196429}$$

$$= 12.61029908$$

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Assignment-03

12.03.2021

Name: Iffat Ara Sanzida
Roll: 344

Given Data-Set including my weight (51kg) is:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
36	37	40	41	41	44	44	45	47

x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{16}	x_{17}	x_{18}
48	48	50	51	51	51	51	51	52

x_{19}	x_{20}	x_{21}	x_{22}	x_{23}	x_{24}	x_{25}	x_{26}	x_{27}
55	55	55	55	58	59	60	60	64

x_{28}	x_{29}	x_{30}	x_{31}	x_{32}	x_{33}	x_{34}	x_{35}	x_{36}
65	67	68	70	75	75	78	78	87

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(a) Construct a stem and leaf plot.

Steam	Leaf
3	6, 7
4	0, 1, 1, 4, 4, 5, 7, 8, 8
5	0, 1, 1, 1, 1, 1, 2, 5, 5, 5, 5, 8, 9
6	0, 0, 4, 5, 7, 8
7	0, 5, 5, 8, 8
8	7

(b) Calculate the sample mean, sample variance and sample standard deviation.

$\Rightarrow$ Sample mean, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

$= \frac{1}{36}\sum_{i=1}^{36} x_i$

$= \frac{1}{36}(36+37+\cdots+78+87)$

$= \frac{2012}{36}$

$= 55.89$

(Ans.)

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~~Sample Variance, $S^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$~~

Sample Variance, $S_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$

$= \frac{1}{36-1} \{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2 + \dots + (x_{36} - \bar{x})^2\}$

$= \frac{5567.5556}{35}$

$= 159.0730171$ (Ans.)

$\therefore$ Sample Standard Deviation, $S_x = \sqrt{S_x^2}$

$= \sqrt{159.0730171}$

$= 12.6124152$ (Ans.)

(c) Construct an ordered Stem & Leaf plot:

Stem	Leaf
3	6, 7
4	0, 1, 1, 4, 4, 5, 7, 8, 8
5	0, 1, 1, 1, 1, 1, 2, 5, 5, 5, 5, 8, 9
6	0, 0, 4, 5, 7, 8
7	0, 5, 5, 8, 8
8	7

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(d) Find the meadian and the first and the third quartiles:

Here, sample size, $n = 36$ which is even.

Median, $Me = \frac{1}{2}\left[x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}\right]$

$= \frac{1}{2}\left[x_{\left(\frac{36}{2}\right)} + x_{\left(\frac{36}{2}+1\right)}\right]$

$= \frac{1}{2}\left[x_{18} + x_{19}\right]$

$= \frac{1}{2}\left[52+55\right]$

$= 53.5$ (Ans.)

Quartiles, $Q_i = \frac{1}{2}\left[x_{\left(\frac{in}{4}\right)} + x_{\left(\frac{in}{4}+1\right)}\right]$

∴ First Quartile, $Q_1 = \frac{1}{2}\left[x_{\left(\frac{1\times 36}{4}\right)} + x_{\left(\frac{1\times 36}{4}+1\right)}\right]$

$= \frac{1}{2}\left[x_9 + x_{10}\right]$

$= \frac{1}{2}\left[47+48\right]$

$= 47.5$ (Ans.)

∴ Third Quartile, $Q_3 = \frac{1}{2}\left[x_{\left(\frac{3\times 36}{4}\right)} + x_{\left(\frac{3\times 36}{4}+1\right)}\right]$

$= \frac{1}{2}\left[x_{27} + x_{28}\right]$

$= \frac{1}{2}\left[64+65\right]$

$= 64.5$ (Ans.)

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(e) Find the interquartile range:

$$IQR = Q_3 - Q_1 = 64.5 - 47.5 = 17$$ (Ans.)

(f) Determine the skewness and kurtosis measures.

$$\text{Skewness: } \gamma_1 = \frac{n \sum_{i=1}^{n} (x_i - \bar{x})^3}{(n-1)(n-2)\, s_x^3} \quad \text{for } n > 2$$

$$= \frac{36 \sum_{i=1}^{36} (x_i - \bar{x})^3}{(36-1)(36-2)\,(12.6124152)^3}$$

$$= \frac{36 \times 40695.34272}{35 \times 34 \times 2006.29494}$$

$$= 0.6136284288$$ (Ans.)

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Kurtosis: $\gamma_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \frac{\sum_{i=1}^{n}(x_i-\bar{x})^4}{S_x^4} - 3$

for $n > 3$

$$= \frac{36(36+1)}{(36-1)(36-2)(36-3)} \frac{\sum_{i=1}^{36}(x_i-\bar{x})^4}{(12.6124152)^4} - 3$$

$$= \frac{36 \times 37}{35 \times 34 \times 33} \times \frac{2287889.741}{25304.22479} - 3$$

$$= 0.0667$$

(Ans.)

(g) Co-efficient of Variation, CV = ?

$$\Rightarrow CV = \frac{S_x}{\bar{x}} \times 100\%$$

$$= \frac{12.6124152}{55.89} \times 100\%$$

$$= 22.56649705\%$$

(Ans.)

Here,

~~S_x~~

Standard deviation, $S_x = 12.6124152$

Mean, $\bar{x} = 55.89$

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(h) Mean Deviation, MD(Mean)

$$= \frac{1}{n} \sum_{i=1}^{n} |x_i - \bar{x}|$$

$$= \frac{1}{36} \sum_{i=1}^{36} |x_i - 55.89|$$

$$= \frac{363.12}{36}$$

$$= 10.086$$

(Ans.)

Assignment 04

18.03.2021

Name: Iffat Ara Sanzida
Roll: 344

8.10

An article in Technometrics presents the following data on motor fuel octane ratings of several blends of gasoline:

88.5	87.7	83.4	86.7	87.5	91.5	88.6	100.3
95.6	93.3	94.7	91.1	91.0	94.2	87.8	89.9
88.3	87.6	84.3	86.7	88.2	90.8	88.3	98.8
94.2	92.7	93.2	91.0	90.3	93.4	88.5	90.1
89.2	88.3	85.3	87.9	88.6	90.9	89.0	96.1
93.3	91.8	92.3	90.4	90.1	93.0	88.7	89.9
89.8	89.6	87.4	88.4	88.9	91.2	89.3	94.4
92.7	91.8	91.6	90.4	91.1	92.6	89.8	90.6
91.1	90.4	89.3	89.7	90.3	91.6	90.5	93.7
92.7	92.2	92.2	91.2	91.0	92.2	90.0	90.7

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(a) Construct a stem and leaf plot.

⇒ At first, we have to sort the given data in ascending order in order to have an ordered stem and leaf plot.

Sorted data are given below:

83.4	84.3	85.3	86.7	86.7	87.4	87.5	87.6
87.7	87.8	87.9	88.2	88.3	88.3	88.3	88.4
88.5	88.5	88.6	88.6	88.7	88.9	89.0	89.2
89.3	89.3	89.6	89.7	89.8	89.8	89.9	89.9
90.0	90.1	90.1	90.3	90.3	90.4	90.4	90.4
90.5	90.6	90.7	90.8	90.9	91.0	91.0	91.0
91.1	91.1	91.1	91.2	91.2	91.5	91.6	91.6
91.8	91.8	92.2	92.2	92.2	92.3	92.6	92.7
92.7	92.7	93.0	93.2	93.3	93.3	93.4	93.7
94.2	94.2	94.4	94.7	95.6	96.1	98.8	100.3

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Stem	Leaf
82	4
84	3
85	3
86	7, 7
87	4, 5, 6, 7, 8, 9
88	2, 3, 3, 3, 4, 5, 5, 6, 6, 7, 9
89	0, 2, 3, 3, 6, 7, 8, 8, 9, 9
90	0, 1, 1, 3, 3, 4, 4, 4, 5, 6, 7, 8, 9
91	0, 0, 0, 1, 1, 1, 2, 2, 5, 6, 6, 8, 8
92	2, 2, 2, 3, 6, 7, 7, 7.
93	0, 2, 3, 3, 4, 7, 8
94	2, 2, 4, 7
95	6
96	1
98	8
100	3

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(b) Construct a frequency distribution and histogram.

⇒ Here, Minimum = 83.4, Maximum = 100.3,

$R = 100.3 - 83.4 = 16.9 \cong 17$, $n = 80$

$K = 1 + 3.332 \log_{10}(80) = 7.34 \cong 8$

$h = \frac{R}{K} = \frac{17}{8} = 2.125 \cong 3$

Cl	x_i	f_i	c.f	$f_i x_i$	$f_i(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^3$	$f_i(x_i - \bar{x})^4$
83-86	84.5	3	3	253.5	112.089	-685.145	4187.98
86-89	87.5	19	22	1662.5	184.07	-572.92	1783.21
89-92	90.5	36	58	3258	0.46	-0.05	0.005
92-95	93.5	18	76	1683	150.08	433.356	1251.327
95-98	96.5	2	78	193	69.33	408.18	2403.16
98-101	99.5	2	80	199	157.98	1404.05	12478.49
Total		80		7249	674.01	987.47	22104.17

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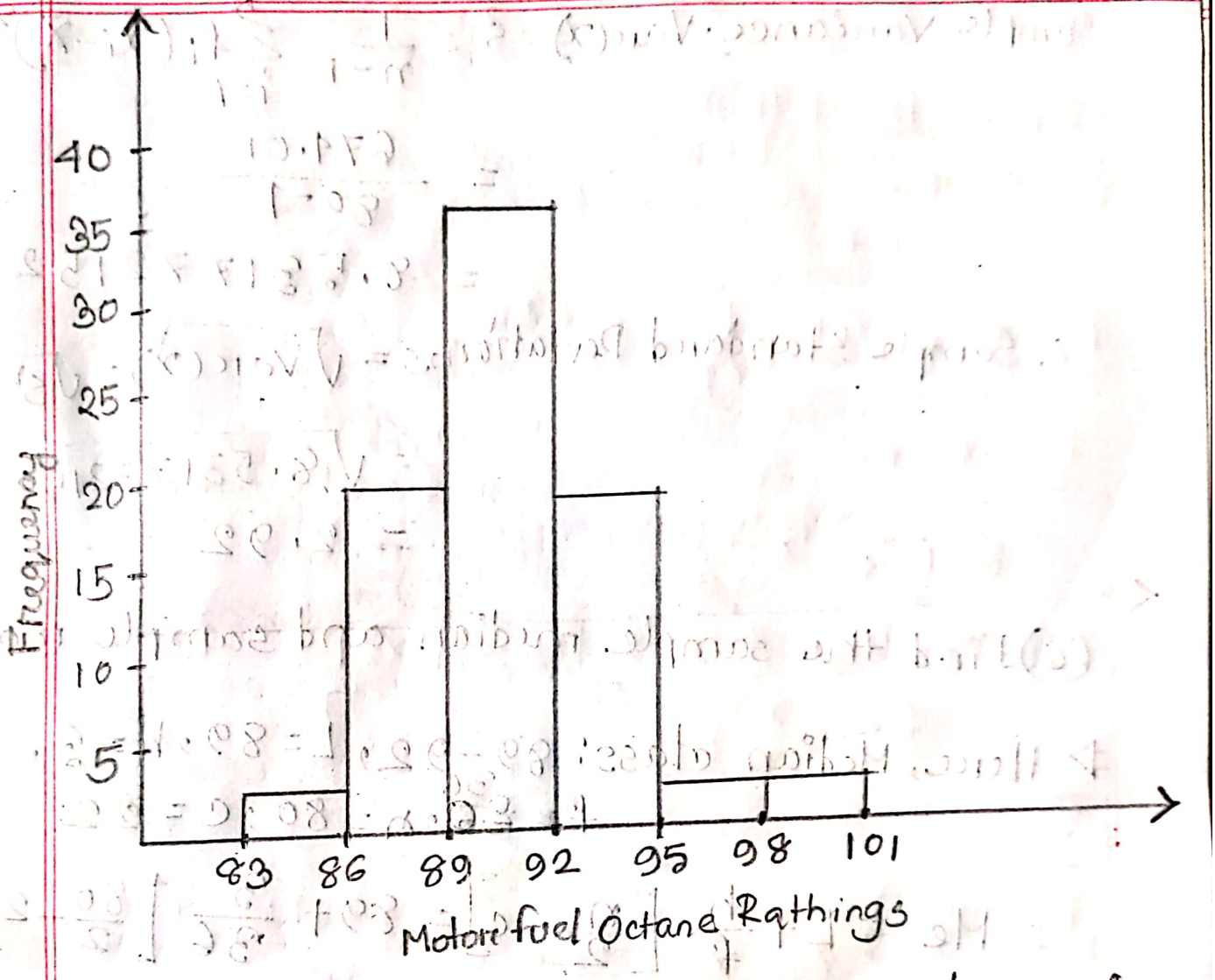

(c) Calculate the sample mean, sample variance and sample standard deviation.

⇒ Using the frequency table from (b),

Sample mean, $\bar{x} = \frac{1}{n}\sum_{i=1}^{k} f_i x_i = \frac{1}{80} \times 7249$

$= 90.6125$

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Sample Variance, $Var(x) = s^2 = \frac{1}{n-1} \sum_{i=1}^{k} f_i (x_i - \bar{x})^2$

$= \frac{674.01}{80-1}$

$= 8.531772152$

$\therefore$ Sample standard Deviation, $s = \sqrt{Var(x)} = \sqrt{s^2}$

$= \sqrt{8.531772152}$

$= 2.92$

(d) Find the sample median and sample mode.

▷ Here, Median class: 89-92, $L = 89, h = 3,$

$f = \cancel{80}\ 36, n = 80, c = 22$

$\therefore Me = L + \frac{h}{f}\left[\frac{n}{2} - c\right] = 89 + \frac{3}{36}\left[\frac{80}{2} - 22\right]$

$= 90.5$

Here, Modal class: 89-92, $L = 89, h = 3, f_p = 19, f_m = 36,$

$f_a = 18$

$Mo = L + \frac{h(f_m - f_p)}{(f_m - f_p) + (f_m - f_a)} = 89 + \frac{3 \times (36-19)}{(36-19)+(36-18)}$

$= 90.457$

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(e) Determine the skewness and kurtosis measures.

⇒ Let assume that the sample mean is the unbiased estimate of population mean. Here,

$$\mu_2 = \frac{1}{80}\sum_{i=1}^{k} f_i(x_i-\bar{x})^2 = \frac{1}{80}\times 674.01 = 8.425$$

$$\mu_3 = \frac{1}{80}\sum_{i=1}^{k} f_i(x_i-\bar{x})^3 = \frac{1}{80}\times 987.47 = 12.34$$

$$\mu_4 = \frac{1}{80}\sum_{i=1}^{k} f_i(x_i-\bar{x})^4 = \frac{1}{80}\times 22104.17 = 276.302$$

Skewness:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(12.34)^2}{(8.425)^3} = 0.254$$

and $\gamma_1 = \sqrt{\beta_1} = 0.503 > 0$

Kurtosis:

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{276.302}{(8.425)^2} = 3.89$$

$$\gamma_2 = \beta_2 - 3 = 0.89 > 0$$

$\because \gamma_1 > 0$ and $\mu_3 > 0$, distribution is positively skewe

$\gamma_2 > 0$ and $\beta_2 > 3$, distribution is Lepto kurtic.

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8.19

Show the following:

(a) That $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$

$\Rightarrow \bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$

for $i = 1$,

$(x_1 - \bar{x}) = x_1 - \frac{x_1 + x_2 + \cdots + x_n}{n}$

for $i = 2$,

$(x_2 - \bar{x}) = x_2 - \frac{x_1 + x_2 + \cdots + x_n}{n}$

$\vdots$

for $i = n$,

$(x_n - \bar{x}) = x_n - \frac{x_1 + x_2 + \cdots + x_n}{n}$

$\therefore \sum_{i=1}^{n}(x_i - \bar{x}) = \left(x_1 - \frac{x_1 + x_2 + \cdots + x_n}{n}\right) + \left(x_2 - \frac{x_1 + x_2 + \cdots + x_n}{n}\right)$

$+ \cdots\cdots + \left(x_n - \frac{x_1 + x_2 + \cdots + x_n}{n}\right)$

$= (x_1 + x_2 + \cdots + x_n) - n\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right)$

$= (x_1 + x_2 + \cdots + x_n) - (x_1 + x_2 + \cdots + x_n)$

$= 0$ [Showed]

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(b) Show that, $\sum_{i=1}^{n}(x_i-\bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\cdot\bar{x}^2$

⇒ At first, we know,

Mean, $\bar{x} = \frac{x_1+x_2+\cdots+x_n}{n}$

$\therefore (x_1+x_2+\cdots+x_n) = n\cdot\bar{x}$ ——— (1)

$$L.H.S = \sum_{i=1}^{n}(x_i-\bar{x})^2$$

$$= \sum_{i=1}^{n}(x_i^2 - 2x_i\bar{x} + \bar{x}^2)$$

$$= \sum_{i=1}^{n} x_i^2 - 2\bar{x}\sum_{i=1}^{n} x_i + \sum_{i=1}^{n}\bar{x}^2$$

$$= \sum_{i=1}^{n} x_i^2 - 2\bar{x}\cdot n\bar{x} + n\bar{x}^2 \quad [\text{from } (1)]$$

$$= \sum_{i=1}^{n} x_i^2 - 2n\bar{x}^2 + n\bar{x}^2$$

$$= \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

$$= \sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n}\bar{x}^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

$$= R.H.S$$

[Showed]

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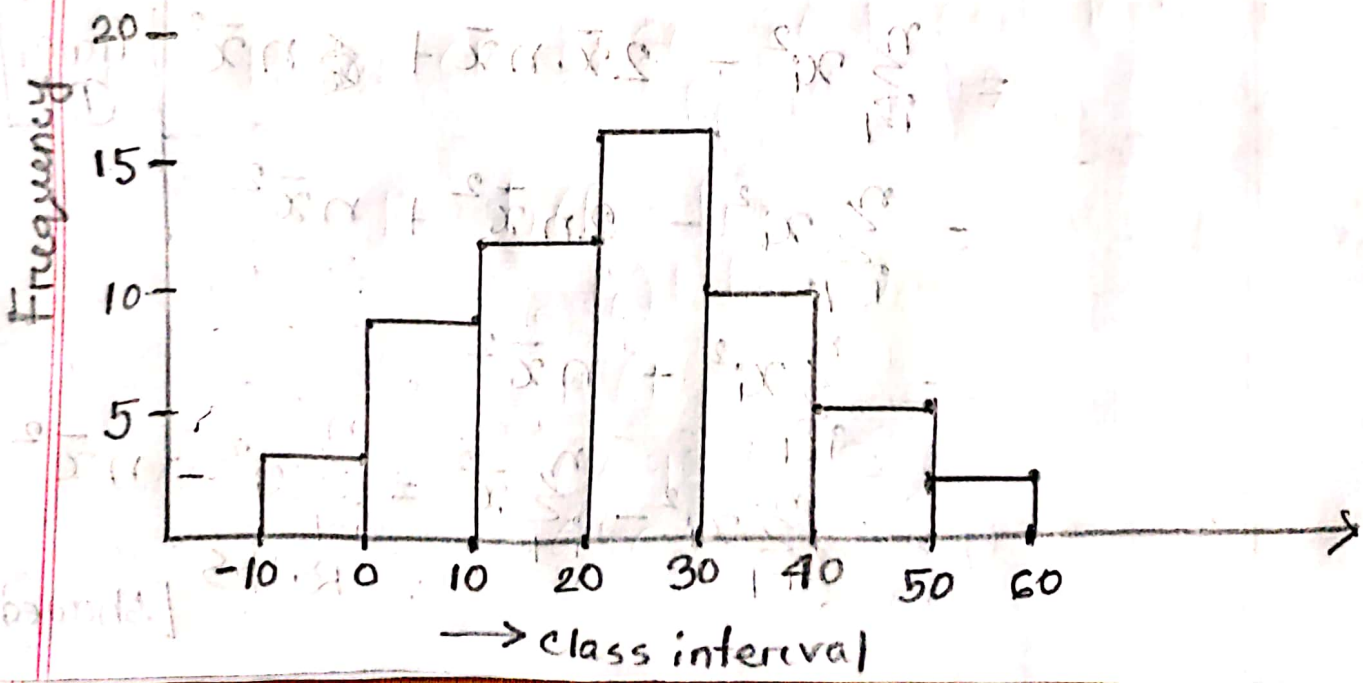
8.33

Class Interval	Frequency
$-10 \le x < 0$	3
$0 \le x < 10$	8
$10 \le x < 20$	12
$20 \le x < 30$	16
$30 \le x < 40$	9
$40 \le x < 50$	4
$50 \le x < 60$	2

i) Construct a frequency distribution and histogram.

⇒



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ii) Calculate sample mean, sample variance and sample standard deviation:

Class	x_i	f_i	C.f	$f_i x_i$	$f_i(x_i-\bar{x})^2$	$f_i(x_i-\bar{x})^3$	$f_i(x_i-\bar{x})^4$
-10 - 0	-5	3	3	-15	2253.43	-61759.75	1692849.468
0 - 10	5	8	11	40	2424.03	-42195.09	734489.03
10 - 20	15	12	23	180	658.36	-4876.47	36120.013
20 - 30	25	16	39	400	107.58	278.95	723.32
30 - 40	35	9	48	315	1427.25	17973.35	226338.39
40 - 50	45	4	52	180	2041.77	46129.71	1042208.54
50 - 60	55	2	54	110	2124.61	69247.42	2256981.16
Total		54		1210	11037.03	24798.12	5989510.821

Sample mean, $\bar{x} = \frac{1}{n}\sum_{i=1}^{k} f_i x_i$

$= \frac{1}{54} \times 1210$

$= 22.407$

Sample variance, $s^2 = \frac{1}{n-1}\sum_{i=1}^{k} f_i(x_i-\bar{x})^2$

$= \frac{1}{54-1} \times 11037.03$

$= 208.2458491$

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∴ sample standard deviation, $S = \sqrt{S^2}$

$= \sqrt{208.2458491}$

$= 14.43$

(iii) Find the sample Median and sample Mode.

⇒ Here, Median class : 20-30, $L = 20, h = 10, f = 16$

$n = 54, c = 23$

$\therefore Me = L + \frac{h}{f}\left[\frac{n}{2} - c\right]$

$= 20 + \frac{10}{16}\left[\frac{54}{2} - 23\right]$

$= 20 + \frac{10}{16}\left[27 - 23\right]$

$= 22.5$

Here, Modal Class : 20 - 30, $L = 20, h = 10, f_m = 16,$

$f_p = 12, f_a = 9$

$\therefore M_o = L + \frac{h(f_m - f_p)}{(f_m - f_p) + (f_m - f_a)}$

$= 20 + \frac{10(16-12)}{(16-12)+(16-9)}$

$= 23.636$

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(iv) Determine the skewness and kurtosis measure.

⇒ Let assume that the sample mean is the unbiased estimate of population mean. Here,

$$\mu_2 = \frac{1}{54}\sum_{i=1}^{k} f_i(x_i-\bar{x})^2 = \frac{11037.03}{54} = 204.389$$

$$\mu_3 = \frac{1}{54}\sum_{i=1}^{k} f_i(x_i-\bar{x})^3 = \frac{24798.12}{54} = 459.224$$

$$\mu_4 = \frac{1}{54}\sum_{i=1}^{k} f_i(x_i-\bar{x})^4 = \frac{5989510.821}{54} = 110916.8671$$

Skewness:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(459.224)^2}{(204.389)^3} = 0.0246$$

$$\gamma_1 = \sqrt{\beta_1} = \sqrt{0.0246} = 0.157$$

Kurtosis:

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{110916.8671}{(204.389)^2} = 2.655$$

$$\gamma_2 = \beta_2 - 3 = 2.655 - 3 = -0.345$$

$\because \gamma_1 > 0$ and $\mu_3 > 0$, positively skewed distribution

$\gamma_2 < 0$ and $\beta_2 < 3$, the distribution is platy kurtic.

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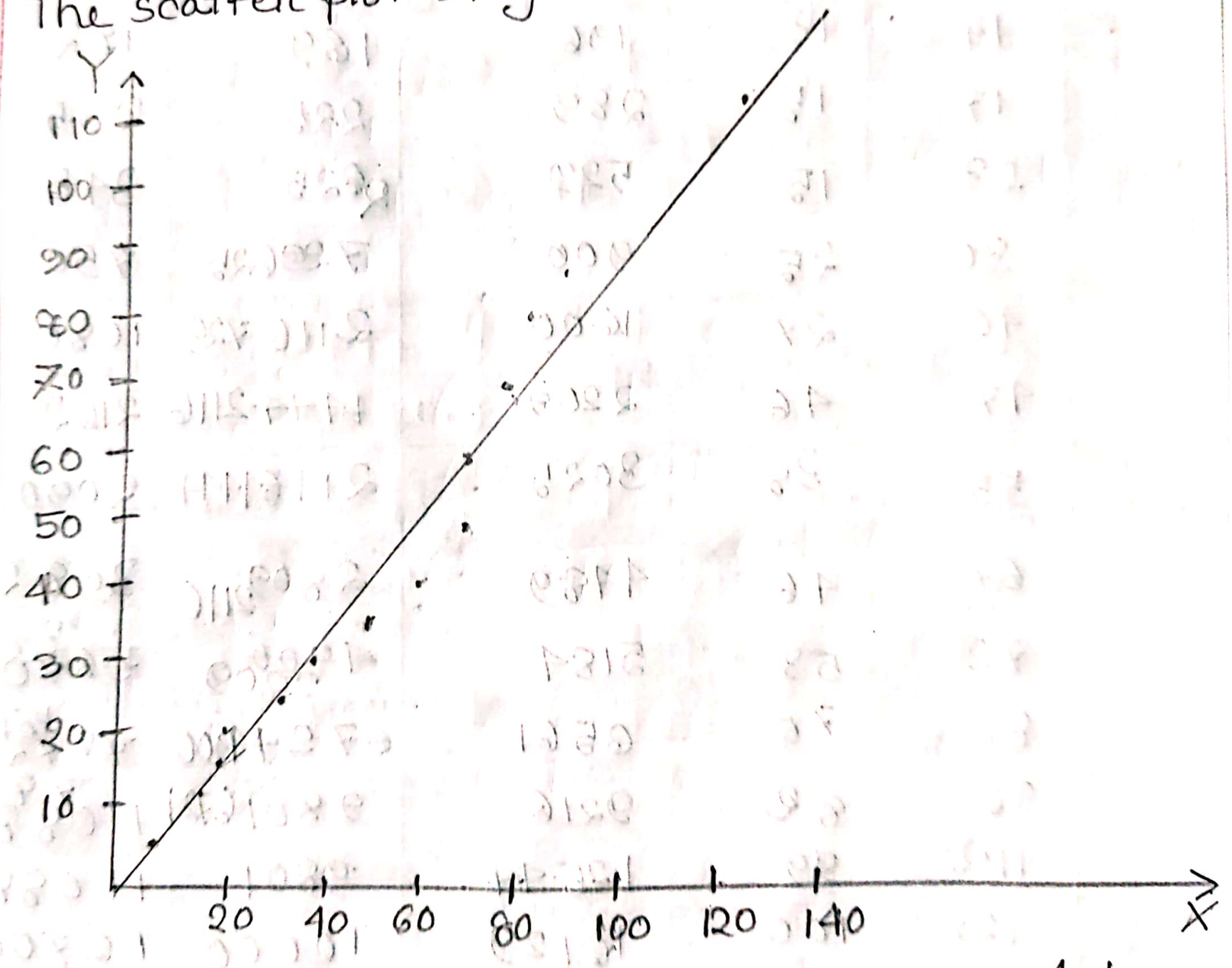
Assignment No-05 30.03.2021

Name: Iffat Arra Sanzida
Roll : 341

16(a) Simple linear regression in a way to describe a relationship between two variables through an equation of a straight line, called line of best fit.

The scatter plot of given data:



We can see from scatter plot that data support the use of the simple linear regression model.

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(b) $\bar{X} = 53.2$ and $\bar{Y} = 42.867$

X	Y	X^2	Y^2	XY
5	4	25	16	20
12	10	144	100	120
14	13	196	169	182
17	15	289	225	255
23	15	529	2025	345
30	25	900	729625	750
40	27	1600	2116 729	1080
47	46	2209	1444 2116	2162
55	38	3025	2116 1444	2090
67	46	4489	2809 2116	3082
72	53	5184	4906 2809	5670 3816
81	70	6561	6724 4900	7872 5670
96	82	9216	9801 6724	11088 7872
112	99	12544	9801	11088
127	100	16129	10000	12700
$\Sigma X_i = 798$	$\Sigma Y_i = 643$	$\Sigma X_i^2 = 63040$	$\Sigma Y^2 = 41999$	$\Sigma XY = 51232$

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Let's, the equation for line of best fit:

$\hat{Y} = \hat{B}_0 + \hat{B}_1 x$ ——— (i)

$\hat{B}_0 = \bar{Y} - \hat{B}_1 \bar{x}$ ——— (ii)

$$\hat{B}_1 = \frac{SP(XY)}{SS(X)} = \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{Y}}{\sum x_i^2 - n\bar{x}^2}$$

$$= \frac{51232 - (15 \times 53.2 \times 42.8677)}{63040 - (15 \times 53.2 \times 53.2)}$$

$$= 0.82697$$

from (ii), $\hat{B}_0 = 42.8667 - (0.82697 \times 53.2)$

$= -1.1282$

Slope of the population regression line, $\hat{B}_1 = 0.82697$ and intercept $\hat{B}_0 = -1.1282$.

(c) We can get the population regression line

$\hat{Y} = -1.1282 + 0.82697 x$

when, $x = 50\ m^3$

$\bar{Y} = -1.1282 + (0.82697 \times 50)$

$= 40.2203\ m^3$

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(d) Standard deviation,

$$\sigma = \sqrt{\frac{SSE}{n-2}}$$

$$= \sqrt{\frac{\sum Y_i^2 - \hat{\beta}_0 \sum Y_i - \hat{\beta}_1 \sum X_i Y_i}{n-2}}$$

$$= \sqrt{\frac{41999 - (-1.1282 \times 643) - (0.8297 \times 51232)}{15-2}}$$

$$= 5.2414$$ (Ans.)

(e) $r^2 = 1 - \frac{SSE}{SST} = 1 - \frac{357.07}{14435.7}$

$$= 0.9753$$

So, 97.53% of the observed variation in runoff value can be attributed to the simple linear regression relationship between runoff and rainfall.

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19(a)

X	Y	X^2	Y^2	XY
100	150	10000	22500	15000
125	140	15625	19600	17500
125	180	15625	32400	22500
150	210	22500	44100	31500
150	190	22500	36100	28500
200	320	40000	102400	64000
200	280	40000	78400	56000
250	400	62500	160000	100000
250	430	62500	184900	107500
300	440	90000	193600	132000
300	390	90000	152100	117000
350	600	122500	360000	210000
400	610	160000	372100	244000
400	670	160000	448900	268000
$\Sigma X_i = 3030$	$\Sigma Y_i = 5010$	$\Sigma X_i^2 = 913750$	$\Sigma Y_i^2 = 2207100$	$\Sigma X_i Y_i = 1413500$

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$$\bar{X} = \frac{3030}{14} = 216.43$$

$$\bar{Y} = \frac{5010}{14} = 357.86$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \quad \text{——} (1)$$

$$\hat{\beta}_1 = \frac{SP(XY)}{SS(X)} = \frac{\sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2}$$

$$= \frac{1413500 - (14 \times 216.43 \times 357.86)}{913750 - \{14 \times (216.43)^2\}}$$

$$= 1.2761$$

$$\hat{\beta}_0 = 357.86 - (1.2761 \times 216.43)$$

$$= 81.682$$

$$\bar{Y} = 81.682 + 1.276X$$

(b) When $X = 225$

$$\bar{Y} = 81.682 + (1.2761 \times 225)$$

$$= 368.8$$

(Ans.)

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(c) Estimated expected change $= -50\hat{\beta}_1$

$= -50 \times 1.2761$

$= -63.805$

(d) No, the value of 500 is outside the range of x values, for which observations were available (the danger of extrapolation).

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Assignment 06 | Name: Iffat Ara Sanjida | 21.04.21
Roll: 344 (1)

Answer to the Exercise No-01

Conditional Probablity:

For any two events A and B with $P(B) > 0$, the conditional probablity of A given that B has occured is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Tree Diagram:

Tree diagram is a tool in the fields of general Mathematics, probablity and statistics that helps calculate the number of possible outcomes of an event or problem and ~~probid~~ provide a visual representation of probablities.

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Bayes Theorem:

Bayes theorem describes the probablity of occurrance of an event related to any condition.

Statement:

Let $E_1, E_2, \ldots, E_n$ be a set of events associated with a sample space S, where all events have nonzero probability of occurrance and they form a partition of S. Let A be any event associated with S, then according to Bayes theorem,

$$P(E_i|A) = \frac{P(E_i) \cdot P(A|E_i)}{\sum_{k=1}^{n} P(E_k) \cdot P(A|E_k)}$$

for any $k = 1, 2, \ldots, n$

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~~Proof:~~

Proof:

According to the conditional probability formula:

$$P(E_i|A) = \frac{P(E_i \cap A)}{P(A)} \quad \text{——— (I)}$$

Using the multiplication rule of probability,

$$P(E_i \cap A) = P(E_i) \cdot P(A|E_i) \quad \text{——— (II)}$$

Using total probability theorem,

$$P(A) = \sum_{k=1}^{n} P(E_k) \cdot P(A|E_k) \quad \text{——— (III)}$$

From (I);

$$P(E_i|A) = \frac{P(E_i) \cdot P(A|E_i)}{\sum_{k=1}^{n} P(E_k) \cdot P(A|E_k)}$$

[Proved]

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Given, $P(B|A) > P(B)$

if $P(B'|A) \not< P(B')$ then,

assume that, $P(B'|A) \geq P(B')$

$\Rightarrow 1 - P(B|A) \geq 1 - P(B)$

$\Rightarrow -P(B|A) \geq -P(B)$

$\Rightarrow P(B|A) \leq P(B)$

which contradicts with our initial condition.

So, $P(B'|A) \geq P(B')$ is not possible.

$\therefore P(B|A) < P(B')$

[showed]

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Ans. to the Exercise No-02

(i) The probablity that the purchaser selected a 21-ft^3 refrigerator is,

$$0.17 + 0.13 = 0.30$$

The probablity that the purchaser selected a refrigerator with an ice-maker is,

$$0.12 + 0.17 + 0.24 = 0.53$$

(ii) Given that the purchaser selected a model with an icemaker, the probablity that it was a 21-ft^3 is, $\frac{0.17}{0.12 + 0.17 + 0.24} = 0.321$

On the other hand, the probablity of selecting a 21 ft^3 model unconditionally is $(0.17 + 0.13) = 0.30$ which is less than the conditional probablity.

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(iii) Given that the purchaser selected a model with at least 20-ft³ of space, the probablity that the selected refrigerator did not have an ice maker is,

$$\frac{0.13+0.16}{0.17+0.13+0.24+0.16} = 0.4143$$

Answer to the Ex-No-03

a) Let's denote the events as,

S: She will receive a strong recommendation.
M: She will receive a moderately good recommendation
W: She will receive a weak recommendation.
R: She will receive the job offer.

Here, $P(R|S) = 0.80$, $P(R'|S) = 0.20$
$P(R|M) = 0.40$, $P(R'|M) = 0.60$
$P(R|W) = 0.10$, $P(R'|W) = 0.90$

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$P(S) = 0.7$, $P(M) = 0.2$, $P(W) = 0.1$

(a) As the probablity of receiving strong, moderate and weak recommendation sum up to 1, we can surely say that, she must have to get the job via either three type of recommendations.

$$P(R) = P(S) \cdot P(R|S) + P(M) \cdot P(R|M) + P(W) \cdot P(R|W)$$

$$= 0.7 \times 0.8 + 0.2 \times 0.4 + 0.1 \times 0.1$$

$$= 0.65$$

(b) $$P(S|R) = \frac{P(S \cap R)}{P(R)} = \frac{P(S) \cdot P(R|S)}{P(R)}$$

$$= \frac{0.7 \times 0.8}{0.65}$$

$$= 0.862$$

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$$P(M|R) = \frac{P(M \cap R)}{P(R)} = \frac{P(M) \cdot P(R|M)}{P(R)}$$

$$= \frac{0.2 \times 0.10}{0.65}$$

$$= 0.123$$

$$P(W|R) = \frac{P(W \cap R)}{P(R)} = \frac{P(W) \times P(R|W)}{P(R)}$$

$$= \frac{0.1 \times 0.1}{0.65}$$

$$= 0.01538$$

~~(c) $P(S|R')$~~

(c) $P(R') = 1 - P(R) = 1 - 0.65 = 0.35$

$$\therefore P(S|R') = \frac{P(S \cap R')}{P(R')} = \frac{P(S) \cdot P(R'|S)}{P(R')}$$

$$= \frac{0.7 \times 0.2}{0.35}$$

$$= 0.4$$

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$$P(M|R') = \frac{P(M \cap R')}{P(R')} = \frac{P(M) \cdot P(R'|M)}{P(R')}$$

$$= \frac{0.2 \times 0.6}{0.35}$$

$$= 0.343$$

$$P(W|R') = \frac{P(W \cap R')}{P(R')} = \frac{P(W) \cdot P(R'|W)}{P(R')}$$

$$= \frac{0.1 \times 0.9}{0.35}$$

$$= 0.257$$

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Assignment 07

21.06.2021

Name: Iffat Arca Sanzida
ID : 344

Exercise 2.9

The manager of a man's clothing store is concerned over the inventory of suits, which is currently 30 (all sizes). The number of suits sold from now ~~on~~ to the end of the season is distributed as,

$$P_X(x) = \frac{e^{-20}20^x}{x!}, \quad x = 0, 1, 2, \cdots$$
$$= 0 \qquad \text{otherwise}$$

Find the probablity that he will have suits left over at the season's end.

Ans.

Let X be the number of suits sold from now to the end of the season.

P.M.F of X is given by

$$P(X = x) = \frac{e^{-20}20^x}{x!} \quad x = 0, 1, 2, \cdots$$

P.M.F of the Poisson Distribution,

$$P(X=x) = \frac{e^{-\lambda}\lambda^{x}}{x!} \quad x = 0, 1, 2, \ldots$$

Comparing both $\lambda = 20$

Probablity that he will have suits left over at the season's end is nothing but number of suits sold is less than or equal 29,

$$P(x \leq 29) = \sum_{x=0}^{29} \frac{e^{-20}\,20^{x}}{x!}$$

There are 29 terms and calculating each probablity would be a tedious job, instead, we can use normal approximation to binomial distribution.

Since, $\lambda > 10$, the normal distribution is a good approximation if an appropriate continuity correction is performed.

The mean would be,

$\mu = \lambda = 20$

standard deviation would be

$\sigma = \sqrt{\lambda} = \sqrt{20} = 4.4721$

We need to add a continuity correction factor as we are approximating a discrete probability distribution by a continuous probability distribution.

- If $P(X = n)$ use $P(n-0.5 < x < n+0.5)$
- If $P(X > n)$ use $P(X > n+0.5)$
- If $P(X \leq n)$ use $P(X < n+0.5)$
- If $P(X < n)$ use $P(X < n-0.5)$
- If $P(X \geq n)$ use $P(X > n-0.5)$

Hence,

$P(x \leq 29) = P(X < 29.5)$

$= P\left(\frac{X-\mu}{\sigma} < \frac{29.5-20}{4.4721}\right)$

$= P(Z < 2.12)$ [$Z = \frac{X-\mu}{\sigma}$ follows standard normal dist]

from standard normal table,
$P(Z < 2.12) = 0.9830$, hence
$P(X \leq 29) = 0.9830$
Probablity that he will have suits left over at the season's end is 0.9830.

Exercise - 2.10

A random variable X has a CDF of the form,

$$F_X(x) = 1 - \left(\frac{1}{2}\right)^{x+1}, \quad x = 0, 1, 2, \cdots$$
$$= 0 \qquad x < 0$$

(a) Find the probablity function for X
(b) Find $P_X(0 < X \leq 8)$

Q) We know that, Probability density function or probability function tells us the probability of certain event occuring. The random variable X is the event that we're considering. On the other hand, cumulative distribution function (CDF) tells us the summation of probabilities. If CDF of a function is F(t), t is always less than or equal to X.

$F(t) = \leq Pxi$

Given that, CDF, $F_x(x) = 1-(\frac{1}{2})^{x+1}$, $x=0,1,2$

So, x is a discrete random variable:

x	$F(x) = 1-(\frac{1}{2})^{x+1}$	P(x)
0	$F(0) = 1-(\frac{1}{2})^1 = \frac{1}{2}$	$P(0) = \frac{1}{2}$
1	$F(1) = 1-(\frac{1}{2})^2 = \frac{3}{4}$	$P(1) = \frac{1}{4}$
2	$F(2) = 1-(\frac{1}{2})^3 = \frac{7}{8}$	$P(2) = \frac{1}{8}$
3	$F(3) = 1-(\frac{1}{2})^4 = \frac{15}{16}$	$P(3) = \frac{1}{16}$
4	$F(4) = 1-(\frac{1}{2})^5 = \frac{31}{32}$	$P(4) = \frac{1}{32}$
5	$F(5) = 1-(\frac{1}{2})^6 = \frac{63}{64}$	$P(5) = \frac{1}{64}$
6	$F(6) = 1-(\frac{1}{2})^7 = \frac{127}{128}$	$P(6) = \frac{1}{128}$
7	$F(7) = 1-(\frac{1}{2})^8 = \frac{255}{256}$	$P(7) = \frac{1}{256}$
8	$F(8) = 1-(\frac{1}{2})^9 = \frac{511}{512}$	$P(8) = \frac{1}{512}$

(b) $P_x(0<x\leq 8)$

$$P_x(0<x\leq 8) = P(1)+P(2)+P(3)+P(4)+P(5)+P(6)+P(7)+P(8)$$

$$= \frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\frac{1}{64}+\frac{1}{128}+\frac{1}{256}+\frac{1}{512}$$

$$= \frac{255}{512} \text{ (Ans.)}$$

2.11

Consider the following probability density function:

$$f_x(x) = kx, \quad 0\leq x<2$$
$$= k(4-x), \quad 2\leq x\leq 4$$
$$= 0, \quad \text{otherwise}$$

(a) Find the value of k for which f is a probability density function.

(b) find the mean and variance of X.

(c) find the cumulative distribution function.

(a) We know, for continuous random variable, f will be PDf if $\int_{-\infty}^{\infty} f(x)dx = 1$

$$\int_0^2 f(x)dx + \int_2^4 f(x)dx = 1$$

$$\Rightarrow \int_0^2 Kx\,dx + \int_2^4 K(4-x)dx = 1$$

$$\Rightarrow K\left[\frac{x^2}{2}\right]_0^2 + K\left[4x - \frac{x^2}{2}\right]_2^4 = 1$$

$$\Rightarrow 2K + K\left[16 - \frac{16}{2} - 8 + \frac{4}{2}\right] = 1$$

$$\Rightarrow 2K + K[16 - 8 - 8 + 2] = 1$$

$$\Rightarrow 2K + 2K = 1$$

$$\Rightarrow 4K = 1$$

$$\Rightarrow K = \frac{1}{4}$$

∴ f is a probability density function if $K = \frac{1}{4}$.

(b) $K = \frac{1}{4}$

$\therefore f_x(x) = \frac{x}{4}, \quad 0 \le x < 2$

$= \frac{1}{4}(4-x), \quad 2 \le x \le 4$

$$\text{Mean}, \mu = E(x) = \int_{-\infty}^{\infty} x \cdot f_x(x)\,dx$$

$$= \int_{-\infty}^{0} x \cdot 0\,dx + \int_{0}^{2} x \cdot \frac{x}{4}\,dx + \int_{2}^{4} x \cdot \frac{1}{4}(4-x)\,dx + \int_{4}^{\infty} 0\,dx$$

$$= 0 + \frac{1}{4}\int_{0}^{2} x^2\,dx + \frac{1}{4}\int_{2}^{4} (4x - x^2)\,dx + 0$$

$$= \frac{1}{4}\left[\frac{x^3}{3}\right]_0^2 + \frac{1}{4}\left[\frac{4x^2}{2} - \frac{x^3}{3}\right]_2^4$$

$$= \frac{1}{4} \times \frac{8}{3} + \frac{1}{4}\left[32 - \frac{64}{3} - 8 + \frac{8}{3}\right]$$

$$= \frac{2}{3} + \frac{1}{4}\,8 - \frac{16}{3} - 1 + \frac{2}{3}$$

$$= \frac{2 + 24 - 16 - 3 + 2}{3}$$

$$= \frac{2}{3} = \frac{28-22}{3} = \frac{6}{3} = 2$$

$= \frac{1}{4} \times \frac{8}{3} + \frac{1}{4}\left[\left(\frac{96-64}{3}\right) - (\right.$

$\therefore$ Mean $= 2$

Variance, $Var(X) = E(X^2) - \{E(X)\}^2$

Now, for $E(X^2) = \int_{-\infty}^{\infty} x^2 f_x(x)\,dx$

$= \int_{-\infty}^{0} x^2 f_x(x)\,dx + \int_{0}^{2} x^2 f_x(x)\,dx + \int_{2}^{4} x^2 f_x(x)\,dx + \int_{4}^{\infty} x^2 f_x(x)\,dx$

$= 0 + \frac{1}{4}\int_{0}^{2} x^3\,dx + \frac{1}{4}\int_{2}^{4} (4x^2 - x^3)\,dx + 0$

$= \frac{1}{4}\left[\frac{x^4}{4}\right]_0^2 + \frac{1}{4}\left[\frac{4x^3}{3} - \frac{x^4}{4}\right]_2^4$

$= \frac{1}{4} \times 4 + \frac{1}{4}\left[\frac{256}{3} - \frac{256}{4} - \frac{32}{3} + \frac{16}{4}\right]$

$= 1 + \frac{64}{3} - 16 - \frac{8}{3} + 1$

$= \frac{3 + 64 - 48 - 8 + 3}{3}$

$= \frac{14}{3}$

$$\therefore \text{Var}(X) = E(X^2) - \{E(X)\}^2$$

$$= \frac{14}{3} - 4$$

$$= \frac{2}{3}$$

(c) Given that,

$$f_x(x) = \begin{cases} \frac{1}{4}x, & 0 \le x < 2 \\ \frac{1}{4}(4-x), & 2 \le x \le 4 \\ 0, & \text{otherwise} \end{cases}$$

For,

$$f_x(x) = \frac{1}{4}x, \quad 0 \le x \le 2$$

$$F(x) = \int_0^x \frac{1}{4} t \, dt = \frac{1}{4}\left[\frac{t^2}{2}\right]_0^x$$

$$= \frac{1}{4} \cdot \frac{x^2}{2} = \frac{x^2}{8}$$

for, $f_x(x) = \frac{1}{4}(4-x)$, $2 \le x \le 4$

$$f(2) = \frac{2^2}{8} = \frac{1}{2}$$

$$F(x) = \frac{1}{2} + \int_2^x \frac{1}{4}(4-t)\,dt$$

$$= \frac{1}{2} + \frac{1}{4}\left[4t - \frac{t^2}{2}\right]_2^x$$

$$= \frac{1}{2} + \frac{1}{4}\left[\left(4x - \frac{x^2}{2}\right) - 8\right]$$

$$= \frac{1}{2} + x - \frac{x^2}{8} - 2$$

$$= x - \frac{x^2}{8} + \frac{3}{2}$$

$$F(x) = \frac{x^2}{8}, \quad 0 \le x \le 2$$

$$= x - \frac{x^2}{8} - \frac{3}{2}, \quad 2 \le x \le 4$$

$$= 0, \quad \text{otherwise}$$

(Ans:)

Exercise-2.21

A continuous random variable X has a density function

$$f_x(x) = \frac{2x}{9}, \quad 0 < x < 3$$
$$= 0 \quad \text{otherwise}$$

(a) Develop the CDF for X.

(b) Find the mean of X and the variance of X.

(c) Find σ_x

(d) find a value ~~of~~ m such that $P_x(X \geq m) = P_x(X \leq m)$. This is called the median of X.

(a) $F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt$

$\therefore F(x) = \int_{-\infty}^{0} f(t)\,dt + \int_{0}^{3} f(t)\,dt + \int_{3}^{x} f(t)\,dt$

$= 0 + \int_{0}^{3} \frac{2t}{9}\,dt + \int_{3}^{x} \frac{2t}{9}\,dt$

$= \frac{2}{9}\left[t^2/2\right]_0^3 + \frac{2}{9}\left[t^2/2\right]_3^x$

$= \frac{2}{9}\cdot\frac{9}{2} + \frac{2}{9}\left[\frac{x^2}{2} - \frac{9}{2}\right]$

$= 1 + \frac{2}{9}\cdot\frac{x^2}{2} - \frac{2}{9}\cdot\frac{9}{2}$

$= \frac{x^2}{9}$

$\therefore F(x) = \frac{x^2}{9}, \quad 0 < x < 3$

$= 0$ otherwise

(b) Mean, $\mu = E(X) = \int_{-\infty}^{\infty} x f_x(x) dx$

$= \int_{-\infty}^{0} x \cdot 0 \, dx + \int_{0}^{3} x \cdot \frac{2x}{9} dx + \int_{3}^{\infty} x \cdot 0 \, dx$

$= 0 + \frac{2}{9} \int_{0}^{3} x^2 dx + 0$

$= \frac{2}{9} \left[\frac{x^3}{3} \right]_0^3$

$= \frac{2}{9} \cdot \left(\frac{3^3}{3} - 0 \right)$

$= 2$

$\therefore$ Mean $= 2$ (Ans.)

Variance, $Var(X) = E(X^2) - \{E(X)\}^2$

for, $E(X^2) = \int_{-\infty}^{\infty} x^2 f_x(x) dx$

$= \int_{-\infty}^{0} x^2 \cdot 0 \, dx + \int_{0}^{3} x^2 \cdot \frac{2x}{9} dx + \int_{3}^{\infty} x^2 \cdot 0 \, dx$

$= 0 + \frac{2}{9} \int_{0}^{3} x^3 dx + 0$

$= \frac{2}{9} \cdot \left[\frac{x^4}{4}\right]_0^3 = \frac{2}{9}\left(\frac{3^4}{4} - 0\right)$

$= \frac{2}{9} \times \frac{81}{4} = \frac{9}{2}$

$\text{Var}(x) = \frac{9}{2} - 2^2$

$= \frac{1}{2}$ (Am.)

(c) $\mu_3' = \int_{-\infty}^{\infty} x^3 f_x(x)\, dx$

$= \int_{-\infty}^{\infty} x^3 \cdot 0\, dx + \int_0^3 x^3 \frac{2x}{9} dx + \int_3^{\infty} 0 \cdot x^3 dx$

$= 0 + \frac{2}{9}\int_0^3 x^4 dx + 0$

$= \frac{2}{9}\left[\frac{x^5}{5}\right]_0^3$

$= \frac{2}{9} \times \frac{3^5}{5} = \frac{54}{5}$

(Am.)

(d) The condition for median,

$$\int_0^m f_x(x)\,dx = \frac{1}{2} \text{ and } \int_m^3 f_x(x)\,dx = \frac{1}{2}$$

if,

$$\int_0^m \frac{2x}{9}\,dx = \frac{1}{2}$$

$$\Rightarrow \frac{2}{9}\int_0^m x\,dx = \frac{1}{2}$$

$$\Rightarrow \frac{2}{9}\left[\frac{x^2}{2}\right]_0^m = \frac{1}{2}$$

$$\Rightarrow \frac{2}{9}\left[\frac{m^2}{2} - 0\right] = \frac{1}{2}$$

$$\Rightarrow \frac{m^2}{9} = \frac{1}{2}$$

$$\Rightarrow m = \pm\sqrt{\frac{9}{2}}$$

$$= \pm\frac{3}{\sqrt{2}}$$

if,

$$\int_m^3 \frac{2x}{9}\,dx = \frac{1}{2}$$

$$\Rightarrow \frac{2}{9}\left[\frac{x^2}{2}\right]_m^3 = \frac{1}{2}$$

$$\Rightarrow \frac{2}{9}\times\frac{9}{2} - \frac{2}{9}\times\frac{m^2}{2} = \frac{1}{2}$$

$$\Rightarrow 1 - \frac{m^2}{9} = \frac{1}{2}$$

$$\Rightarrow \frac{m^2}{9} = \frac{1}{2}$$

$$\Rightarrow m = \pm\frac{3}{\sqrt{2}}$$

Exercise 2.24

Find the cumulative distribution function associated with $f_x(x) = \frac{1}{\sigma\pi} \frac{1}{\left\{1+\frac{(x-\mu)^2}{\sigma^2}\right\}}$, $-\infty < x < \infty$

$$CDF = \int_{-\infty}^{\infty} \frac{1}{\sigma x} \frac{1}{1+\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

Suppose, $\frac{x-\mu}{\sigma} = y$

$\Rightarrow \frac{x}{\sigma} - \frac{\mu}{\sigma} = y$

$\Rightarrow \frac{dx}{\sigma} - 0 = dy$

$\Rightarrow dx = \sigma dy$

$x \to \infty, \quad y \to \infty$

$x \to -\infty, \quad y \to -\infty$

$$\text{CDF} = \int_{-\infty}^{\infty} \frac{1}{\pi\sigma} \cdot \frac{\sigma \, dy}{1+y^2}$$

$$= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{1}{1+y^2} dy$$

$$= \frac{1}{\pi} \left[\tan^{-1} y \right]_{-\infty}^{\infty}$$

$$= \frac{1}{\pi} \left[\tan^{-1}(\infty) - \tan^{-1}(-\infty) \right]$$

$$= \frac{1}{\pi} \left\{ \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \right\}$$

$$= \frac{1}{\pi} \cdot \pi$$

$$= 1$$

(Ans.)

Assignment On Exponential, Gamma and Weibull Distribution.

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6-13

The time to failure of a television tube is estimated to be exponentially distributed with a mean of 3 years. A company offers insurance on these tubes for the first year of usage. On what percentage of policies will they have to pay a claim?

⇒ Given, $\mu = \frac{1}{\lambda} = 3$

$\therefore \lambda = \frac{1}{3}$

then, $P(x \leq 1) = \int_0^1 \frac{1}{3} e^{-\frac{x}{3}} dx$

$= \frac{1}{3}(-3)\left[e^{-\frac{x}{3}}\right]_0^1$

$= -1\left(e^{-\frac{1}{3}} - 1\right)$

$= 1 - e^{-\frac{1}{3}}$

$= 0.28346$

∴ They will have to pay a claim 28.346% of policies.

6-14

Is there an exponential distribution that satisfies the following condition?

$$P(X \le 2) = \frac{2}{3} P(X \le 3)$$

If so, find the value of λ.

⇒ Exponential distribution function,

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & ; x \ge 0 \\ 0 & ; \text{otherwise} \end{cases}$$

then, $P(X \le 2) = \int_0^2 \lambda e^{-\lambda x} dx$

$$= \lambda \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^2$$

$$= -(e^{-2\lambda} - 1)$$

$$= 1 - e^{-2\lambda}$$

$$P(X \le 3) = \int_0^3 \lambda e^{-\lambda x} dx$$

$$= \lambda \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^3$$

$$= -(e^{-3\lambda} - 1)$$

$$= 1 - e^{-3\lambda}$$

Given,

$$1-e^{-2\lambda} = \frac{2}{3}(1-e^{-3\lambda})$$

$$\Rightarrow 3-3e^{-2\lambda} = 2-2e^{-3\lambda}$$

$$\Rightarrow 1-3e^{-2\lambda}+2e^{-3\lambda} = 0$$

$$\Rightarrow 2(e^{-\lambda})^3 - 3(e^{-\lambda})^2 + 1 = 0$$

$$\Rightarrow 2y^3 - 3y^2 + 1 = 0 \quad [\text{Let, } e^{-\lambda} = y]$$

$$\Rightarrow (y-1)^2(2y+1) = 0$$

$(y-1)^2 = 0$

$\Rightarrow y = 1$

$\Rightarrow e^{-\lambda} = 1$

$\Rightarrow -\lambda = 0$

$\Rightarrow \lambda = 0$

$(2y+1) = 0$

$\Rightarrow 2y = -1$

$\Rightarrow y = -\frac{1}{2}$

$\Rightarrow e^{-\lambda} = -\frac{1}{2}$

6-15

Two manufacturing processes are under consideration. The per-unit cost for process I is C, while for process II it is 3C. Products from both processes have exponential time to failure densities with mean rates of 25^{-1} failures per hour and 35^{-1} failures per hour from I and II respectively. If a product fails before 15 hours it must be replaced at a cost of 2 dollars. Which process ~~wou~~ would you recommend?

⇒ According to the question,

$C_I = C$; $x \geq 15$

$= C + 2$; $x < 15$

$C_{II} = 3C$ ∵, $x \geq 15$

$= 3C + 2$; $x < 15$

The expected costs are,

$$E(C_I) = 2\int_0^{15} \frac{1}{25} \cdot e^{-\frac{x}{25}} dx + C\int_0^{15} \frac{1}{25} \cdot e^{-\frac{x}{25}} dx + C\int_{15}^{\infty} \frac{1}{25} \cdot e^{-\frac{x}{25}} dx$$

$$= \frac{c}{25}(-25)\cdot\left[e^{-\frac{x}{25}}\right]_0^{15} + \frac{2}{25}(-25)\left[e^{-\frac{x}{25}}\right]_0^{15}$$

$$+ \frac{c}{25}(-25)\left[e^{-\frac{x}{25}}\right]_{15}^{\infty}$$

$$= -c\left(e^{-\frac{3}{5}}-1\right) - 2\left(e^{-\frac{3}{5}}-1\right) + (-c)\left(-e^{-\frac{3}{5}}\right)$$

$$= -ce^{-\frac{3}{5}} + c - 2e^{-\frac{3}{5}} + 2 + ce^{-\frac{3}{5}}$$

$$= c + 2\left(1-e^{-\frac{3}{5}}\right)$$

$$= c + 0.4512\,z$$

$$E(c_1) = 3c\int_0^{15}\frac{1}{35}e^{-\frac{x}{35}}dx + 2\int_0^{15}\frac{1}{35}e^{-\frac{x}{35}}dx$$

$$+ 3c\int_{15}^{\infty}\frac{1}{35}e^{-\frac{x}{35}}dx$$

$$= \frac{3c}{35}(-35)\left[e^{-\frac{x}{35}}\right]_0^{15} + \frac{2}{35}(-35)\left[e^{-\frac{x}{35}}\right]_0^{15}$$

$$+ \frac{3c}{35}(-35)\left[e^{-\frac{x}{35}}\right]_{15}^{\infty}$$

$$= -3c\left(e^{-\frac{3}{7}}-1\right) + 2\left(e^{-\frac{3}{7}}-1\right)$$

$$+ (-3c)\left(-e^{-\frac{3}{7}}\right)$$

$$= -3ce^{-\frac{3}{7}} + 3c - 2e^{-\frac{3}{7}} + 2 + 3ce^{-\frac{3}{7}}$$

$$= 3c + 2\left(1-e^{-\frac{3}{7}}\right)$$

$$= 3c + 0.3485\,z$$

As, $E(C_{II}) > E(C_I)$, I will recommend process I.

6-16

A transistor has an exponential time to failure distribution with a mean time to failure of 20000 hours. The transistor has already lasted 20000 hours in a particular application. What is the probability that the transistor fails by 30000 hours?

$\Rightarrow$ Given, $\mu = \frac{1}{\lambda} = 20000$

$\Rightarrow \lambda = \frac{1}{20000}$

$$P(X \leq 30000) = \int_0^{30000} \frac{1}{20000} e^{-\frac{x}{20000}} dx$$

$$= \frac{1}{20000} (-20000) \left[e^{-\frac{x}{20000}}\right]_0^{30000}$$

$$= -\left(e^{-\frac{3}{2}} - 1\right)$$

$$= 1 - e^{-\frac{3}{2}}$$

$$= 0.7769$$

(Ans.)

6-34

Graph the Weibull distribution with ~~$x=0$~~ $\gamma=0$, $\delta=1$ and $\beta = 1, 2, 3, 1$

$\Rightarrow$ The probablity density function for Weibull distribution,

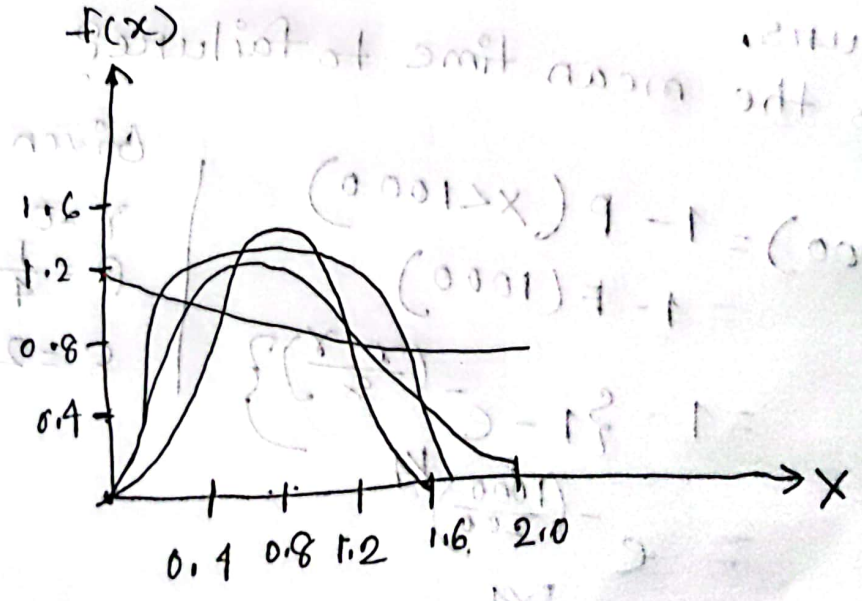
$$f(x) = \frac{\beta}{\delta}\left(\frac{x-\gamma}{\delta}\right)^{\beta-1} \cdot e^{-\left(\frac{x-\gamma}{\delta}\right)^{\beta}} \quad ; x \geq \gamma$$

$$= 0 \quad ; \text{otherwise}$$

$\gamma\ (-\infty < \gamma < \infty) \rightarrow$ the location parameter.

$\delta > 0 \rightarrow$ the scale parameter

for, $\gamma = 0$, $\delta = 1$



Weibull densities for $\gamma = 0$, $\delta = 1$ and $\beta = 1, 2, 3$ and 4 using density function, we get,

$\beta = 1$, $f(x) = e^{-x}$

$\beta = 2$, $f(x) = 2x e^{-2x}$

$\beta = 3$, $f(x) = 3x^2 e^{-3x}$

$\beta = 4$, $f(x) = 4x^3 \cdot e^{-4x}$

$\gamma = 0$
$\delta = 1$

6-35

The time to failure density for a small computer system has a weibull density with $\gamma = 0$, $\beta = \frac{1}{4}$ and $\delta = 200$

(a) What is fraction of these units will survive to 1000 hours.

(b) What is the mean time to failure?

Given,
$\gamma = 0$
$\beta = \frac{1}{4}$
$\delta = 2000$

(a) $P(X \geq 1000) = 1 - P(X < 1000)$

$= 1 - F(1000)$

$= 1 - \left\{1 - e^{-\left(\frac{x-\gamma}{\delta}\right)^{\beta}}\right\}$

$= e^{-\left(\frac{1000}{200}\right)^{1/4}}$

$= e^{-5/4}$

$= 0.2865$ (Ans.)

(b) $\gamma = 0$

$\delta = 200$

$\beta = \frac{1}{5}$

$E(X) = \gamma + \delta \Gamma(1 + \frac{1}{\beta})$

$= 0 + 200 \times \Gamma 5$

$= 200 \times 4!$

$= 200 \times 24$

$= 4800$ hours (Ans.)